Fundamental Theorem of Matrix Product States
A Formalization Blueprint

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Chapter 1

Introduction

This blueprint describes the fundamental theorem for periodic, translation-invariant matrix product vectors. The basic object is a family of matrices $\{A^i\}$, viewed as a tensor that generates vectors $|V^{(N)}(A)\rangle$ on chains of length N.

Chapters 2 and 3 describe the algebraic core. Chapter 2 introduces word evaluation, MPV coefficients, gauge equivalence, injectivity, blocking, overlap, and the block-injective canonical-form data. Chapter 3 proves the single-block fundamental theorem by a purely algebraic route through trace pairings, linear extension, simplicity of matrix algebras, and Skolem–Noether.

The later chapters develop the spectral and canonical-form machinery needed for the full multi-block theorem. Chapter 4 studies positive maps and quantum channels. Chapters 5 and 6 supply Perron–Frobenius theory and overlap decay. Chapters 7 and 8 connect blocking, primitivity, irreducibility, and the normal/canonical-form reductions. Chapters 9 and 10 carry out block permutation and BNT theory; Chapter 11 states the final assembly.

Chapter 2

Matrix Product Vectors

Fix a physical dimension d and a bond dimension D. This chapter studies the tensors that generate matrix product vector (MPV) families. All tensors in this blueprint are translation-invariant with periodic boundary conditions (PBC).

2.1 Basic definitions

Definition 2.1 (MPS tensor)**.** A (translation-invariant, PBC) *MPS tensor* with physical dimension d and bond dimension D is a collection of matrices

$$\{A^i\}_{i=0}^{d-1}, \qquad A^i \in M_D(\mathbb{C}),$$

indexed by a physical index $i \in \{0, \dots, d-1\}$. Such a tensor defines an MPV family.

Definition 2.2 (Word evaluation)**.** Given a word $w = (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$, the *word evaluation* is the matrix product

$$A^w \ := \ A^{i_1} A^{i_2} \cdots A^{i_L} \ \in \ M_D(\mathbb{C}),$$

with the convention that the empty word evaluates to the identity: $A^{\emptyset} = \mathbb{1}_D$.

Lemma 2.3 (Word evaluation respects concatenation)**.** *For any words w_1, w_2, $A^{w_1 \cdot w_2} = A^{w_1} A^{w_2}$.*

Proof. Immediate from the definition of A^w. □

Definition 2.4 (Matrix product vector)**.** The *matrix product vector* (MPV) of a tensor A at system size N is the vector

$$|V^{(N)}(A)\rangle = \sum_{i_1,\dots,i_N=0}^{d-1} \operatorname{tr}(A^{i_1} A^{i_2} \cdots A^{i_N}) |i_1, \dots, i_N\rangle \in (\mathbb{C}^d)^{\otimes N}.$$

Equivalently, for a configuration $\sigma = (i_1, \dots, i_N) \in \{0, \dots, d-1\}^N$, we write

$$V^{(N)}(A)_\sigma := \operatorname{tr}(A^{i_1} A^{i_2} \cdots A^{i_N}).$$

The coefficient function $\sigma \mapsto V^{(N)}(A)_\sigma$ gives the components; the displayed ket is the corresponding vector in $(\mathbb{C}^d)^{\otimes N}$. For a general word w, we also write

$$c_w(A) := \operatorname{tr}(A^w),$$

The *MPV family* generated by A is the collection $\mathcal{V}(A) = \{|V^{(N)}(A)\rangle\}_{N \geq 1}$.

Definition 2.5 (Transfer map)**.** The *transfer map* associated to a tensor A is the linear map $\mathcal{E}_A : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$\mathcal{E}_A(X) \;=\; \sum_{i=0}^{d-1} A^i\, X\, (A^i)^\dagger .$$

Remark 2.6. The transfer map is automatically positive: if $X \geq 0$ then $\mathcal{E}_A(X) \geq 0$, because each summand $A^i X (A^i)^\dagger$ is positive semidefinite. The abstract positive-map framework appears in Chapter 4.

2.2 Gauge equivalence and same MPV

Definition 2.7 (Gauge equivalence)**.** Two tensors A and B of the same bond dimension D are *gauge equivalent* if there exists an invertible matrix $X \in \mathrm{GL}_D(\mathbb{C})$ such that

$$B^i \;=\; X\, A^i\, X^{-1} \qquad \text{for all } i \in \{0, \dots, d-1\}.$$

Definition 2.8 (Same MPV)**.** Two tensors A and B (of the same bond dimension) *generate the same MPV family* if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every system size $N \geq 1$.

Definition 2.9 (Same MPV — different bond dimensions)**.** For tensors A and B of possibly different bond dimensions D_1 and D_2, we write $\mathcal{V}(A) = \mathcal{V}(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every $N \geq 1$.

Remark 2.10. Definition 2.8 is the same-bond-dimension specialization of Definition 2.9. We keep both because later results use both the same-dimension and different-dimension forms.

Definition 2.11 (Proportional MPV)**.** Two tensors A and B (of possibly different bond dimensions) generate *proportional MPV families* if for every N there exists $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N\, |V^{(N)}(B)\rangle$.

Lemma 2.12 (Word evaluation under scaling)**.** *Let ζA denote the tensor with matrices $\{\zeta\, A^i\}_{i=0}^{d-1}$. Then for any word w,*

$$(\zeta A)^w \;=\; \zeta^{|w|}\, A^w,$$

where $|w|$ is the length of w.

Proof. Induction on w: the base case gives $\zeta^0 = 1$, and each step picks up one factor of ζ. ☐

Definition 2.13 (Gauge-phase equivalence)**.** Two tensors A and B are *gauge-phase equivalent* if there exist $X \in \mathrm{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C} \setminus \{0\}$ such that

$$B^i \;=\; \zeta\, X\, A^i\, X^{-1} \qquad \text{for all } i.$$

Remark 2.14. After spectral-radius normalization one may restrict ζ to a unit-modulus phase $e^{i\phi}$. We allow an arbitrary nonzero scalar because later canonical-form statements keep the block-scaling factors explicit.

Lemma 2.15 (Word evaluation under conjugation)**.** *If $B^i = X\, A^i\, X^{-1}$ for all i, then for every word w,*

$$B^w \;=\; X\, A^w\, X^{-1}.$$

Proof. Induction on the word length, using $XX^{-1} = \mathbb{1}$. ☐

Theorem 2.16 (Gauge invariance of MPVs)**.** *If A and B are gauge equivalent, then they generate the same MPV family.*

Proof. Let $B^i = XA^iX^{-1}$. By Lemma 2.15, $B^w = X\,A^w\,X^{-1}$ for every word w. Then $\mathrm{tr}(B^w) = \mathrm{tr}(X\,A^w\,X^{-1}) = \mathrm{tr}(A^w)$ by cyclicity of the trace. $\square$

2.3 Injectivity and normality

Definition 2.17 (Injectivity)**.** A tensor A is *injective* if its matrices span the full matrix algebra: $\mathrm{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d-1\} = M_D(\mathbb{C})$.

Definition 2.18 (L-block injectivity)**.** A tensor A is *L-block injective* if $\mathrm{span}_{\mathbb{C}}\{A^{i_1} \cdots A^{i_L} : (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L\} = M_D(\mathbb{C})$. An injective tensor is 1-block injective.

Definition 2.19 (Normal tensor)**.** A tensor A is *normal* [CPGSV21] if there exists a finite L_0 such that A is L_0-block injective.

Remark 2.20. In the block-injective route developed here, "normal" means eventual block injectivity. The NT notion in [CPGSV21] is spectral: it is phrased in terms of irreducibility and the peripheral spectrum of the transfer map. The bridge to that viewpoint is supplied later by Theorem 8.26, Theorem 7.45, and Theorem 7.52.

2.4 Canonical form

Definition 2.21 (Block-injective canonical form)**.** In this blueprint, a *block-injective canonical form* for a tensor consists of:

1. a number of blocks $r \geq 1$,
2. bond dimensions $D_1, \dots, D_r$,
3. injective tensors $\{A_k^i\}_{i=0}^{d-1}$ with $A_k^i \in M_{D_k}(\mathbb{C})$ for each block $k \in \{1, \dots, r\}$,
4. scaling factors $\mu_1, \dots, \mu_r \in \mathbb{C}$.

The associated tensor is the block-diagonal tensor

$$A^i \;=\; \bigoplus_{k=1}^{r} \mu_k\, A_k^i.$$

This is the block-injective version of the canonical-form data; see [PGVWC07] and [CPGSV21].

Remark 2.22. This blueprint uses the block-injective (biCF-style) canonical form: each block is injective in the sense of Definition 2.17. This is stronger than the NT-block canonical form of [CPGSV21], where the blocks are only normal in the paper's spectral sense. The paper-faithful NT/BNT/CFII route is developed later in Chapter 8; the present section records the stronger block-injective package used in the later block-injective assembly.

Definition 2.23 (Block-diagonal tensor from blocks)**.** Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, the associated *block-diagonal tensor* is the tensor of total bond dimension $D = \sum_{k=1}^r D_k$ defined by

$$A^i \;=\; \bigoplus_{k=1}^{r} \mu_k\, A_k^i.$$

Theorem 2.24 (MPV decomposition)**.** *The MPV of a block-diagonal tensor decomposes as*

$$|V^{(N)}(A)\rangle \;=\; \sum_{k=1}^{r} \mu_k^N \, |V^{(N)}(A_k)\rangle.$$

Equivalently, for each configuration σ one has

$$V^{(N)}(A)_\sigma \;=\; \sum_{k=1}^{r} \mu_k^N \, V^{(N)}(A_k)_\sigma.$$

Proof. For each physical index i, the block-diagonal tensor $A^i = \bigoplus_k \mu_k A_k^i$ is a block-diagonal matrix. By Lemma 2.12, the word evaluation of each block picks up a factor $\mu_k^{|w|}$, and the full word evaluation is block-diagonal with blocks $\mu_k^N A_k^w$. Taking the trace of a block-diagonal matrix gives the sum of the block traces: $\mathrm{tr}(A^w) = \sum_k \mu_k^N \,\mathrm{tr}(A_k^w)$. □

2.5 Blocking

Definition 2.25 (Blocked tensor)**.** Given a tensor A and a blocking length L, the *L-blocked tensor* is the tensor with physical index set $\{0, \dots, d-1\}^L$ (hence physical dimension d^L) and the same bond dimension D, whose matrices are indexed by words $(i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$:

$$(A^{[L]})^{(i_1,\dots,i_L)} \;=\; A^{i_1} A^{i_2} \cdots A^{i_L}.$$

Lemma 2.26 (Blocked word evaluation)**.** *If $w = (J_1, \dots, J_m)$ is a word in the blocked alphabet, where each J_k is an L-tuple in $\{0, \dots, d-1\}^L$, then concatenating the block words gives a word $\widetilde{w}$ of length mL in the original alphabet and one has*

$$(A^{[L]})^w = A^{\widetilde{w}}.$$

Proof. Induct on the blocked word w and split off the first block word. Lemma 2.3 turns concatenation of blocks into multiplication of the corresponding evaluations. □

Lemma 2.27 (Blocking preserves MPV coefficients)**.** *For each basis state $\sigma = (\sigma_1, \dots, \sigma_N)$ of the blocked system, where each σ_k is an L-tuple in $\{0, \dots, d-1\}^L$, there exists a basis state $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_{NL})$ of the original system such that*

$$V^{(N)}(A^{[L]})_\sigma \;=\; V^{(NL)}(A)_{\widetilde{\sigma}}.$$

Proof. Concatenate the block words and apply the blocked word evaluation lemma (Lemma 2.26). □

Theorem 2.28 (Blocking preserves same MPV)**.** *If A and B generate the same MPV family, then for any blocking length L, the blocked tensors $A^{[L]}$ and $B^{[L]}$ also generate the same MPV family.*

Proof. By Lemma 2.27, each MPV coefficient of the blocked tensor $A^{[L]}$ equals a coefficient of A at the corresponding concatenated configuration. The same holds for $B^{[L]}$ (with the same concatenated configuration). Since A and B agree on all configurations, so do $A^{[L]}$ and $B^{[L]}$. □

2.6 MPV overlap

Definition 2.29 (MPV as Hilbert-space vector)**.** We regard $|V^{(N)}(A)\rangle$ as an element of $\mathbb{C}^{d^N}$ indexed by configurations $\sigma \in \{0, \dots, d-1\}^N$, with σ-component $V^{(N)}(A)_\sigma$ as in Definition 2.4. This allows us to use the standard Euclidean inner product on $\mathbb{C}^{d^N}$.

Definition 2.30 (MPV inner product)**.** The *MPV inner product* between two tensors A and B at system size N is

$$\langle V^{(N)}(A)|V^{(N)}(B)\rangle \;:=\; \sum_{\sigma\in\{0,\dots,d-1\}^N} \overline{V^{(N)}(A)_\sigma}\, V^{(N)}(B)_\sigma,$$

i.e. the Euclidean inner product on $\mathbb{C}^{d^N}$ (conjugate-linear in the first argument).

Definition 2.31 (MPV overlap)**.** The *MPV overlap* between two tensors A (of bond dimension D_1) and B (of bond dimension D_2) at system size N is

$$O_{AB}(N) \;:=\; \sum_{\sigma\in\{0,\dots,d-1\}^N} V^{(N)}(A)_\sigma\, \overline{V^{(N)}(B)_\sigma}.$$

This is the bilinear physics-oriented overlap used later. By Lemma 2.32, it is the complex conjugate of the Hilbert-space inner product from Definition 2.30.

Lemma 2.32 (Overlap equals conjugate inner product)**.** $O_{AB}(N) = \overline{\langle V^{(N)}(A)|V^{(N)}(B)\rangle}$.

Proof. The overlap sums $V_\sigma\,\overline{W_\sigma}$ while the inner product sums $\overline{V_\sigma}\,W_\sigma$. Complex conjugation swaps these. □

Chapter 3

The Single-Block Fundamental Theorem

This chapter proves that if an injective MPS tensor A generates the same MPV family as another tensor B of the same bond dimension, then A and B are gauge equivalent. The proof is purely algebraic and does not use the later overlap or normal-form reductions. The trace-pairing maps introduced below are auxiliary devices attached to the chosen generating families $\{A^i\}$ and $\{B^i\}$; they serve only to construct the linear map that is later shown to be a gauge transform.

3.1 Trace pairing

Theorem 3.1 (Nondegeneracy of the trace pairing). *For $M \in M_D(\mathbb{C})$, $\mathrm{tr}(MN) = 0$ for all $N \in M_D(\mathbb{C})$ if and only if $M = 0$.*

Proof. Testing against the matrix units E_{ji} (the matrix with 1 in position (j, i) and 0 elsewhere) gives $\mathrm{tr}(E_{ji}M) = M_{ij}$. Hence if $\mathrm{tr}(MN) = 0$ for all N, then every entry of M vanishes. ☐

Lemma 3.2 (Same MPV implies trace agreement). *If A and B generate the same MPV family, then $\mathrm{tr}(A^w) = \mathrm{tr}(B^w)$ for every word w.*

Proof. Viewing the word w as a configuration of length $|w|$, Definition 2.4 gives the corresponding MPV coefficient as the trace coefficient $c_w(A) = \mathrm{tr}(A^w)$. The same-MPV hypothesis gives equality at every system size. ☐

Definition 3.3 (Trace pairing map). For a tensor A, attach to the chosen family $\{A^i\}$ the auxiliary linear map $\Phi_A : M_D(\mathbb{C}) \to \mathbb{C}^d$ defined by

$$\Phi_A(M)_i = \mathrm{tr}(M \cdot A^i).$$

Theorem 3.4 (Injectivity of the trace pairing map). *If A is injective, then $\ker \Phi_A = \{0\}$.*

Proof. If $\Phi_A(M) = 0$, then $\mathrm{tr}(M \cdot A^i) = 0$ for all i. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, this gives $\mathrm{tr}(MN) = 0$ for all N, so $M = 0$ by Theorem 3.1. ☐

Theorem 3.5 (Nonvanishing trace on injective tensors). *If $D \geq 1$, A is injective, A and B generate the same MPV family, and $B^i = 0$ for all i, then we have a contradiction.*

Proof. If $B^i = 0$ for all i, then Lemma 3.2 gives $\mathrm{tr}(A^w) = 0$ for every word w, in particular for all length-1 words. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, the trace functional vanishes identically, contradicting $\mathrm{tr}(\mathbb{1}_D) = D \neq 0$. $\square$

3.2 Linear extension

Theorem 3.6 (Existence and uniqueness of the linear extension). *If A is injective and A, B generate the same MPV family, then there exists a unique linear map $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ satisfying $T(A^i) = B^i$ for all i.*

Proof. Write $\ell_A : \mathbb{C}^d \to M_D(\mathbb{C})$ for the map that sends a coefficient vector c to $\sum_i c_i A^i$ (linear combination with the A^i). Applying Lemma 3.2 to all words of length 2 shows $\Phi_A \circ \ell_A = \Phi_B \circ \ell_B$. Since A is injective, ℓ_A is surjective, giving $\mathrm{range}(\Phi_A) \subseteq \mathrm{range}(\Phi_B)$. By Theorem 3.4, $\ker \Phi_A = \{0\}$, so rank–nullity gives $\dim(\mathrm{range}(\Phi_A)) = \dim(M_D(\mathbb{C}))$. Hence $\dim(\mathrm{range}(\Phi_B)) \geq \dim(M_D(\mathbb{C}))$, and rank–nullity again gives $\ker \Phi_B = \{0\}$. A left inverse g of Φ_B gives $T := g \circ \Phi_A$, and uniqueness follows because the $\{A^i\}$ span $M_D(\mathbb{C})$. $\square$

Theorem 3.7 (Multiplicativity of the linear extension). *If A is injective and A, B generate the same MPV family, and $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is the unique linear map satisfying $T(A^i) = B^i$ for all i, then $T(MN) = T(M)T(N)$ for all $M, N \in M_D(\mathbb{C})$.*

Proof. **Step 1** ($\Phi_B \circ T = \Phi_A$). Applying Lemma 3.2 to words of length 2 gives $\Phi_B(T(A^i)) = \Phi_B(B^i) = \Phi_A(A^i)$ for all i. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $\Phi_B \circ T = \Phi_A$.

Step 2 ($T(A^i A^j) = B^i B^j$). Lemma 3.2 applied to the word (i, j, k) gives $\mathrm{tr}(A^i A^j A^k) = \mathrm{tr}(B^i B^j B^k)$ for all i, j, k. So $\Phi_B(T(A^i A^j))_k = \Phi_A(A^i A^j)_k = \mathrm{tr}(A^i A^j A^k) = \mathrm{tr}(B^i B^j B^k) = \Phi_B(B^i B^j)_k$. The same rank–nullity argument used in the proof of Theorem 3.6 gives $\ker \Phi_B = \{0\}$. Hence $T(A^i A^j) = B^i B^j$.

Step 3 (bilinear extension). For fixed j, the maps $M \mapsto T(M \cdot A^j)$ and $M \mapsto T(M) \cdot B^j$ agree on the spanning set $\{A^i\}$, hence on all of $M_D(\mathbb{C})$. Applying this once more with j replaced by a general matrix gives full multiplicativity. $\square$

3.3 Simplicity and Skolem–Noether

Theorem 3.8 (Simplicity of $M_D(\mathbb{C})$). *A nonzero multiplicative $\mathbb{C}$-linear endomorphism of $M_D(\mathbb{C})$ is bijective.*

Proof. The kernel of a multiplicative linear map is a two-sided ideal. Since $M_D(\mathbb{C})$ is a simple ring (it has no nontrivial two-sided ideals), the kernel is either $\{0\}$ or $M_D(\mathbb{C})$. The latter would force the map to be zero, contradicting the hypothesis. Hence the map is injective, and surjectivity follows from finite-dimensionality. $\square$

Theorem 3.9 (Skolem–Noether). *Every $\mathbb{C}$-algebra automorphism of $M_D(\mathbb{C})$ is inner: for any $f \in \mathrm{Aut}_{\mathbb{C}\text{-}alg}(M_D(\mathbb{C}))$, there exists $X \in \mathrm{GL}_D(\mathbb{C})$ such that $f(M) = XMX^{-1}$ for all M.*

Proof. Via the canonical identification $M_D(\mathbb{C}) \cong \mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$, the automorphism f induces an automorphism of $\mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$. Every algebra automorphism of $\mathrm{End}(V)$ is conjugation by an invertible linear map (this is a standard result in the theory of central simple algebras). Translating back to matrices gives the invertible X. $\square$

Lemma 3.10 (Promotion to algebra homomorphism)**.** *If $T : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a multiplicative surjective $\mathbb{C}$-linear map, then $T(\mathbb{1}) = \mathbb{1}$. Equivalently, the same underlying function defines a $\mathbb{C}$-algebra homomorphism.*

Proof. By surjectivity, there exists X with $T(X) = \mathbb{1}$. Then $T(\mathbb{1}) = T(X) \cdot T(\mathbb{1}) = T(X \cdot \mathbb{1}) = T(X) = \mathbb{1}$, using multiplicativity and $T(X) = \mathbb{1}$. With $T(\mathbb{1}) = \mathbb{1}$ established, T preserves the unit and hence is a $\mathbb{C}$-algebra homomorphism. □

3.4 Assembly

Theorem 3.11 (Single-block Fundamental Theorem)**.** *Let A be an injective MPV tensor and let B be an MPV tensor of the same bond dimension. If A and B generate the same MPV family, then they are gauge equivalent: there exists $X \in \mathrm{GL}_D(\mathbb{C})$ such that $B^i = XA^iX^{-1}$ for all i.*

Proof. By Theorem 3.6, there is a unique linear T with $T(A^i) = B^i$. By Theorem 3.7, T is multiplicative. T is nonzero: if $T = 0$ then $B^i = 0$ for all i, contradicting Theorem 3.5. By Theorem 3.8, T is bijective. By Lemma 3.10, T is a $\mathbb{C}$-algebra automorphism. Theorem 3.9 gives an invertible X with $T(M) = XMX^{-1}$, and evaluating at $M = A^i$ yields $B^i = XA^iX^{-1}$. □

Chapter 4

Quantum Channels and Positive Maps

This chapter develops the theory of positive maps and quantum channels on matrix algebras, following [Wol12].

4.1 Positive maps and channels

Definition 4.1 (Positive map)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *positive* if $E(X) \geq 0$ whenever $X \geq 0$, where $X \geq 0$ means that X is positive semidefinite.

Remark 4.2. We use the Loewner order on matrices: $X \leq Y$ means $Y - X \geq 0$.

Definition 4.3 (Trace-preserving map)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *trace-preserving* if $\mathrm{tr}(E(X)) = \mathrm{tr}(X)$ for all X.

Definition 4.4 (Completely positive map)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *completely positive* (CP) if it admits a *Kraus representation*: there exist operators $\{K_i\}_{i=0}^{r-1}$ with $K_i \in M_D(\mathbb{C})$ such that

$$E(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger \quad \text{for all } X \in M_D(\mathbb{C}).$$

We use the Kraus characterisation because it matches the later Kadison–Schwarz arguments. By Choi's theorem, this is equivalent to $E \otimes \mathrm{id}_n$ being positive for all n.

Theorem 4.5 (CP maps are positive)**.** *Every completely positive map is positive.*

Proof. If $X \geq 0$, then each summand $K_i X K_i^\dagger \geq 0$ (conjugation preserves positive semidefiniteness), so the sum is PSD. □

Definition 4.6 (Quantum channel)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is a *quantum channel* if it is both completely positive and trace-preserving (CPTP), following [Wol12].

Definition 4.7 (Density matrices)**.** The set of *density matrices* in $M_D(\mathbb{C})$ is

$$\mathcal{D}_D = \{\rho \in M_D(\mathbb{C}) : \rho \geq 0, \mathrm{tr}(\rho) = 1\}.$$

Theorem 4.8 (Density matrices are compact)**.** *$\mathcal{D}_D$ is compact (in the entry-wise topology on $M_D(\mathbb{C})$).*

Proof. The PSD cone is closed (preimage of the closed nonnegative cone under continuous quadratic forms), and if $\rho \geq 0$ with $\mathrm{tr}(\rho) = 1$ then each entry satisfies $|\rho_{ij}|^2 \leq \rho_{ii}\rho_{jj} \leq 1$: the diagonal entries are nonnegative and sum to 1, and the off-diagonal bound is the PSD Cauchy–Schwarz inequality. Hence the matrix entries are uniformly bounded, and Heine–Borel gives compactness. □

Theorem 4.9 (Density matrices are convex)**.** *$\mathcal{D}_D$ is convex.*

Proof. Convex combination of PSD matrices is PSD, and trace is linear. □

Theorem 4.10 (Density matrices are nonempty)**.** *For $D \geq 1$, $\mathcal{D}_D \neq \emptyset$.*

Proof. $\frac{1}{D}\mathbb{1}_D$ is a density matrix. □

Theorem 4.11 (Channels preserve density matrices)**.** *If E is a quantum channel, then $E(\mathcal{D}_D) \subseteq \mathcal{D}_D$.*

Proof. Complete positivity implies positivity (Theorem 4.5), so $E(\rho) \geq 0$; trace preservation gives $\mathrm{tr}(E(\rho)) = \mathrm{tr}(\rho) = 1$. □

4.2 Kadison–Schwarz inequality

A Kraus map is the concrete realisation of a completely positive map (Definition 4.4). The Kadison–Schwarz inequality and the multiplicative domain characterisation are proved for Kraus maps, then applied to transfer maps, which are Kraus maps by construction (Remark 4.25).

Definition 4.12 (Kraus map)**.** Given operators $\{K_i\}_{i=0}^{d-1}$ with $K_i \in M_D(\mathbb{C})$, the *Kraus map* is $E(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger$. A linear map is completely positive (Definition 4.4) if and only if it can be written in this form.

Definition 4.13 (Adjoint Kraus map)**.** The *adjoint Kraus map* is $E^*(X) = \sum_{i=0}^{d-1} K_i^\dagger X K_i$. When the $\{K_i\}$ are the matrices of an MPS tensor A, this is the transfer map of the conjugate-transposed family $i \mapsto (A^i)^\dagger$.

Definition 4.14 (Unital Kraus map)**.** A Kraus map is *unital* if $\sum_{i=0}^{d-1} K_i K_i^\dagger = \mathbb{1}$. Equivalently, $E(\mathbb{1}) = \mathbb{1}$. In the later gauge language this is the right-canonical condition.

Definition 4.15 (Trace-preserving Kraus map)**.** A Kraus map is *trace-preserving* if $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$. Equivalently, the adjoint Kraus map is unital. This is the standard MPS normalization condition. In the later gauge language it is the left-canonical condition, so Kadison–Schwarz arguments are often applied to the adjoint map.

Theorem 4.16 (Kadison–Schwarz inequality)**.** *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a* unital *Kraus map (i.e. $\sum_i K_i K_i^\dagger = \mathbb{1}$). Then for all $X \in M_D(\mathbb{C})$,*

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Eq. (5.2)].

Proof. The 2×2 block matrix $\left[\begin{smallmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{smallmatrix}\right]$ is PSD ($= vv^\dagger$ with $v = [X^\dagger, \mathbb{1}]^T$). Applying the unital CP map blockwise preserves positive semidefiniteness (each block $K_i(\cdot)K_i^\dagger$ preserves PSD, and unitality ensures the $(2,2)$-block maps to $\mathbb{1}$). The Schur complement of the $(2,2)$-block $\mathbb{1}$ gives the result. □

Theorem 4.17 (Kadison–Schwarz for the adjoint channel)**.** *If $\sum_i K_i^\dagger K_i = \mathbb{1}$ (trace-preserving), then the adjoint Kraus map satisfies $E^*(X^\dagger X) \geq E^*(X)^\dagger E^*(X)$.*

Proof. The adjoint operators $\{K_i^\dagger\}$ satisfy $\sum_i K_i^\dagger (K_i^\dagger)^\dagger = \sum_i K_i^\dagger K_i = \mathbb{1}$, so they form a unital family. Apply Theorem 4.16 to $\{K_i^\dagger\}$. □

Theorem 4.18 (Hilbert–Schmidt contraction)**.** *If a Kraus map is both unital and trace-preserving, then $\mathrm{tr}(E(X)^\dagger E(X)) \leq \mathrm{tr}(X^\dagger X)$ for all $X \in M_D(\mathbb{C})$.*

Proof. By Theorem 4.16, $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$, so $\mathrm{tr}(E(X^\dagger X)) \geq \mathrm{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\mathrm{tr}(E(X^\dagger X)) = \mathrm{tr}(X^\dagger X)$. □

4.3 Multiplicative domain

Theorem 4.19 (KS gap decomposition)**.** *For a unital Kraus map E, the Kadison–Schwarz gap decomposes as*

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_{i=0}^{d-1} R_i^\dagger R_i,$$

where $R_i := XK_i^\dagger - K_i^\dagger E(X)$.

Proof. Expand the right-hand side $\sum_i (XK_i^\dagger - K_i^\dagger E(X))^\dagger (XK_i^\dagger - K_i^\dagger E(X))$, distribute, and use unitality $\sum_i K_i K_i^\dagger = \mathbb{1}$ to identify the result with the KS gap. □

Theorem 4.20 (KS equality implies Kraus commutation)**.** *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$ for some X, then $XK_i^\dagger = K_i^\dagger E(X)$ for all i.*

Proof. By Theorem 4.19, the KS gap is $\sum_i R_i^\dagger R_i$ with $R_i = XK_i^\dagger - K_i^\dagger E(X)$. If the gap vanishes, $\sum_i R_i^\dagger R_i = 0$. Each summand $R_i^\dagger R_i$ is PSD, so each must be zero, giving $R_i = 0$. □

Theorem 4.21 (KS equality for peripheral eigenvectors)**.** *Let E be a Kraus map that is both unital and trace-preserving. If $E(X) = \mu X$ with $|\mu| = 1$, then the KS gap vanishes: $E(X^\dagger X) = E(X)^\dagger E(X)$.*

Proof. The gap $G := E(X^\dagger X) - E(X)^\dagger E(X)$ is PSD by Theorem 4.16. Its trace satisfies $\mathrm{tr}(G) = \mathrm{tr}(E(X^\dagger X)) - \mathrm{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\mathrm{tr}(E(X^\dagger X)) = \mathrm{tr}(X^\dagger X)$. Using $E(X) = \mu X$ and $|\mu| = 1$, $\mathrm{tr}(E(X)^\dagger E(X)) = |\mu|^2 \,\mathrm{tr}(X^\dagger X) = \mathrm{tr}(X^\dagger X)$. So $\mathrm{tr}(G) = 0$, and a PSD matrix with zero trace must be zero. □

Theorem 4.22 (One-sided multiplicative-domain identity)**.** *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then $E(X^\dagger Y) = E(X)^\dagger E(Y)$ for all $Y \in M_D(\mathbb{C})$.*

Proof. From Theorem 4.20, $XK_i^\dagger = K_i^\dagger E(X)$ for all i. Taking conjugate transposes: $K_i X^\dagger = E(X)^\dagger K_i$. Then

$$\begin{aligned} E(X^\dagger Y) &= \sum_i K_i X^\dagger Y K_i^\dagger \\ &= \sum_i E(X)^\dagger K_i Y K_i^\dagger \\ &= E(X)^\dagger E(Y). \end{aligned}$$

□

4.4 Irreducibility

Definition 4.23 (Irreducible map)**.** A linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ is *irreducible* if whenever P is an orthogonal projection satisfying $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, then $P = 0$ or $P = \mathbb{1}$. This is [Wol12, Thm. 6.2, condition (1)]. The definition applies to any linear map; complete positivity is not required.

Theorem 4.24 (Injectivity implies irreducibility)**.** *If A is an injective MPS tensor, then its transfer map $\mathcal{E}_A$ is irreducible.*

Proof. If P is an invariant projection for $\mathcal{E}_A$, then $(\mathbb{1} - P)A^i P = 0$ for all i. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $(\mathbb{1} - P)MP = 0$ for all M, so $P = 0$ or $P = \mathbb{1}$. □

Remark 4.25 (Transfer map is CP)*.* The transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is completely positive: it is already written in Kraus form, with Kraus operators $\{A^i\}$.

Theorem 4.26 (Transfer map is a channel)**.** *If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, then the transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is a quantum channel (CPTP).*

Proof. Complete positivity is the observation of Remark 4.25. Trace preservation: $\operatorname{tr}(\mathcal{E}_A(X)) = \sum_i \operatorname{tr}(A^i X (A^i)^\dagger) = \operatorname{tr}((\sum_i (A^i)^\dagger A^i)X) = \operatorname{tr}(X)$ by cyclicity and the normalization hypothesis. □

4.5 Cesàro fixed-point theorem

Definition 4.27 (Cesàro mean)**.** The *Cesàro mean* of a linear map E at order N, applied to a starting matrix ρ_0, is

$$\sigma_N = \frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho_0).$$

Theorem 4.28 (Cesàro telescope)**.** $E(\sigma_N) - \sigma_N = \frac{1}{N}(E^N(\rho_0) - \rho_0)$*.*

Proof. Telescoping: $E(\sigma_N) = \frac{1}{N}\sum_{n=1}^{N} E^n(\rho_0)$ and subtracting σ_N cancels all but the first and last terms. □

Theorem 4.29 (Hermitian fixed-point decomposition)**.** *Let E be a quantum channel and $X = E(X)$ a Hermitian fixed point with decomposition $X = P_+ - P_-$ into orthogonal positive and negative parts ($P_\pm \geq 0$, $\operatorname{tr}(P_+P_-) = 0$). Then $E(P_+) = P_+$ and $E(P_-) = P_-$. This is [Wol12, Prop. 6.8].*

Proof. Let Q be the support projection of P_+. Trace preservation and positivity give $\operatorname{tr}(QE(P_-)) = 0$ and $E(P_-) \geq 0$, so $QE(P_-) = 0$. The fixed-point equation $E(P_+) - E(P_-) = P_+ - P_-$ then identifies $E(P_\pm) = P_\pm$. $\square$

Theorem 4.30 (Cesàro fixed-point theorem). *Every quantum channel $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ (with $D \geq 1$) has a nonzero positive semidefinite fixed point: there exists $\rho \geq 0$, $\rho \neq 0$, with $E(\rho) = \rho$.*

Proof. Start with any $\rho_0 \in \mathcal{D}_D$ (Theorem 4.10). Each Cesàro mean σ_N lies in $\mathcal{D}_D$ (by convexity and channel preservation). By compactness (Theorem 4.8), extract a convergent subsequence $\sigma_{N_k} \to \rho$. The telescope identity (Theorem 4.28) gives $\|E(\sigma_N) - \sigma_N\| = \frac{1}{N}\|E^N(\rho_0) - \rho_0\| \to 0$, so $E(\rho) = \rho$ by continuity. In [Wol12, Thm. 6.11], existence is proved via Brouwer's fixed-point theorem; we use the Cesàro argument instead. $\square$

4.6 Fixed-point projection

Definition 4.31 (Fixed-point projection). Given a linear map $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ and a fixed point ρ with $E(\rho) = \rho$ and $\operatorname{tr}(\rho) \neq 0$, the *fixed-point projection* is the rank-one map

$$P(X) = \frac{\operatorname{tr}(X)}{\operatorname{tr}(\rho)}\rho.$$

This projects onto the span of ρ along the kernel of the trace functional. When the fixed point is unique up to scaling, this span is the full fixed-point space.

Theorem 4.32 (Power decomposition). *If E is trace-preserving and $E(\rho) = \rho$, then for $n \geq 1$,*

$$E^n = P + (E - P)^n,$$

where P is the fixed-point projection.

Proof. P is idempotent ($P^2 = P$), the fixed-point relation $E(\rho) = \rho$ gives $E \circ P = P$, and trace preservation gives $P \circ E = P$. These imply $(E-P) \circ P = 0$ and $P \circ (E-P) = 0$, so in the binomial expansion of $(P+(E-P))^n$ all cross terms vanish, leaving $E^n = P^n + (E-P)^n = P + (E-P)^n$. $\square$

4.7 Peripheral spectrum and primitivity

Definition 4.33 (Peripheral spectrum). The *peripheral spectrum* of a bounded linear operator T is the set of spectral values μ with $|\mu|$ equal to the spectral radius of T.

Definition 4.34 (Peripheral eigenvalues). The *peripheral eigenvalues* of a linear map E on a finite-dimensional space are the eigenvalues λ on the unit circle: $|\lambda| = 1$. For a channel, the spectral radius is 1 [Wol12, Prop. 6.1], so these coincide with the eigenvalues of maximal modulus.

Note: The condition $|\lambda| = 1$ is appropriate for channels, since the spectral radius of a channel is 1. For a general linear map with spectral radius $r \neq 1$ the condition would generalise to $|\lambda| = r$.

Theorem 4.35 (1 is a peripheral eigenvalue). *If E has a nonzero fixed point $\rho \neq 0$ with $E(\rho) = \rho$, then 1 is a peripheral eigenvalue of E.*

Proof. ρ is an eigenvector with eigenvalue 1, and $|1| = 1$. $\square$

Remark 4.36 (Peripheral eigenvalues lie on the unit circle). Every peripheral eigenvalue λ satisfies $|\lambda| = 1$. This is simply the defining condition in Definition 4.34.

Theorem 4.37 (Finiteness of peripheral eigenvalues)**.** *On a finite-dimensional space, the set of peripheral eigenvalues is finite.*

Proof. Peripheral eigenvalues are a subset of all eigenvalues, which is a finite set in finite dimensions. □

Theorem 4.38 (Power-stable peripheral eigenvalues are roots of unity)**.** *Let E be a linear endomorphism on a finite-dimensional space. If μ is a peripheral eigenvalue of E and μ^n is an eigenvalue of E for every $n \geq 1$, then μ is a root of unity.*

Proof. Since μ^n is an eigenvalue for every n, the pigeonhole principle on the finite eigenvalue set gives $\mu^a = \mu^b$ for some $a \neq b$, hence $\mu^{|a-b|} = 1$. □

Theorem 4.39 (Peripheral powers imply root of unity)**.** *If the set of peripheral eigenvalues of a linear endomorphism E on a finite-dimensional space is* closed under powers *($\mu \in \mathrm{peripheral}(E)$ and $n \geq 1$ imply $\mu^n \in \mathrm{peripheral}(E)$), then every peripheral eigenvalue is a root of unity.*

Proof. The closure hypothesis provides $\mu^n \in \mathrm{peripheral}(E)$ for all n, which in particular means μ^n is an eigenvalue. Theorem 4.38 then gives $\mu^p = 1$ for some $p > 0$. □

4.7.1 Periodicity removal by powering

Remark 4.40. The results in this subsection are purely spectral: they use only finiteness of a set of complex numbers and the spectral mapping theorem for powers. In particular, they do not rely on complete positivity. For transfer maps, replacing a tensor by its p-blocked tensor replaces $\mathcal{E}_A$ by $\mathcal{E}_A^p$, so this powering step is the operator-theoretic form of blocking.

Lemma 4.41 (Common exponent for a finite set of roots of unity)**.** *Let $s \subseteq \mathbb{C}$ be a finite set. Assume that for every $\mu \in s$ there exists an exponent $p_\mu > 0$ with $\mu^{p_\mu} = 1$. Then there exists $p > 0$ such that $\mu^p = 1$ for all $\mu \in s$.*

Proof. Take p to be the least common multiple of the exponents p_μ. □

Lemma 4.42 (Powering collapses peripheral eigenvalues)**.** *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a linear map, and assume that E has a nonzero fixed point $\rho \neq 0$. Let $p > 0$. If every peripheral eigenvalue μ of E satisfies $\mu^p = 1$, then* $\mathrm{peripheral}(E^p) = \{1\}$.

Proof. The fixed point ρ gives $1 \in \mathrm{peripheral}(E^p)$. For the reverse inclusion, let $\nu \in \mathrm{peripheral}(E^p)$. Spectral mapping gives $\nu = \mu^p$ for some $\mu \in \sigma(E)$. From $|\nu| = 1$ and $p > 0$ we obtain $|\mu| = 1$, hence $\mu \in \mathrm{peripheral}(E)$. The hypothesis $\mu^p = 1$ then forces $\nu = 1$. □

For a normalized MPS tensor, the transfer map is naturally trace-preserving but need not be unital. In general one should not expect a single gauge choice to make it both unital and trace-preserving.

Remark 4.43 (Bi-canonical special case). When a Kraus map happens to be both unital and trace-preserving (the "bi-canonical" case), the Kadison–Schwarz equality at peripheral eigenvectors (Theorem 4.21) shows directly that the peripheral eigenvalues are closed under powers, and hence are roots of unity by Theorem 4.39. This is the simplest route, but it requires both normalizations simultaneously. The more general adjoint-fixed-point formulation below covers the case where only unitality and a positive definite adjoint fixed point are available.

4.7.2 Peripheral closure via adjoint fixed point

The following results replace the trace-preservation hypothesis by the existence of a positive definite fixed point for the *adjoint* Kraus map, matching the weighted-trace argument in [CPGSV17, Appendix A].

Lemma 4.44 (Peripheral closure via adjoint fixed point)**.** *Let* $E(X) = \sum_i K_i X K_i^\dagger$ *be a Kraus map that is unital (*$\sum_i K_i K_i^\dagger = \mathbb{1}$*), and assume:*

1. *the adjoint Kraus map* $E^*(X) = \sum_i K_i^\dagger X K_i$ *has a positive definite fixed point* $\rho > 0$*, and*
2. E *is irreducible.*

Then $\mathrm{peripheral}(E)$ *is closed under powers: if* $\mu \in \mathrm{peripheral}(E)$ *then* $\mu^n \in \mathrm{peripheral}(E)$ *for every* $n \in \mathbb{N}$*.*

Proof. Let $E(X) = \mu X$ with $|\mu| = 1$. The weighted Kadison–Schwarz equality (using the adjoint fixed point ρ in place of trace preservation) shows $E(X^\dagger X) = E(X)^\dagger E(X)$. Hence $X^\dagger X$ is a PSD fixed point, which is positive definite by irreducibility (Theorem 5.3). So X is invertible, and multiplicative-domain powers give $E(X^n) = \mu^n X^n$, so μ^n is peripheral. □

Theorem 4.45 (Roots of unity via adjoint fixed point)**.** *Let* $E(X) = \sum_i K_i X K_i^\dagger$ *be a Kraus map that is unital, and assume that the adjoint Kraus map* $E^*(X) = \sum_i K_i^\dagger X K_i$ *has a positive definite fixed point and that* E *is irreducible. Then every peripheral eigenvalue of* E *is a root of unity.*

Proof. Combine the closure-under-powers result (Lemma 4.44) with Theorem 4.39. □

Remark 4.46. Lemma 4.44 and Theorem 4.45 cover the bi-canonical (unital + TP) case as well: trace preservation gives $E^*(\mathbb{1}) = \mathbb{1}$, which is a positive definite fixed point of the adjoint. The adjoint-fixed-point formulation matches the argument in [CPGSV17, Appendix A], which uses a faithful invariant state rather than bi-canonicality.

Definition 4.47 (Channel period)**.** The *period* of a channel E is the number of peripheral eigenvalues (counted without multiplicity): $p(E) := |\mathrm{peripheral}(E)|$.

Definition 4.48 (Primitive map)**.** A linear map is *primitive* if its only peripheral eigenvalue is $\lambda = 1$, i.e. $\mathrm{peripheral}(E) = \{1\}$. For channels this corresponds to [Wol12, Thm. 6.7]. This definition applies to any linear endomorphism, not only to channels.

Theorem 4.49 (Primitivity equals period one)**.** *A linear map with a nonzero fixed point is primitive if and only if its period is* 1*.*

Proof. If $\mathrm{peripheral}(E) = \{1\}$, then $|\mathrm{peripheral}(E)| = 1$. Conversely, if $|\mathrm{peripheral}(E)| = 1$, then since $1 \in \mathrm{peripheral}(E)$ (Theorem 4.35), the only element is 1. □

Theorem 4.50 (Primitive maps have a spectral gap on the complement)**.** *Let* $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ *be trace-preserving, and let* $\rho \neq 0$ *satisfy* $E(\rho) = \rho$ *and* $\mathrm{tr}(\rho) \neq 0$*. Let*

$$P(X) = \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)}\,\rho$$

be the fixed-point projection associated to ρ*. Assume that:*

1. *E is primitive;*

2. *every eigenvalue μ of E satisfies $|\mu| \leq 1$;*

3. *whenever $E(X) = X$ and $\operatorname{tr}(X) = 0$, one has $X = 0$.*

Then every eigenvalue ν of $E - P$ satisfies $|\nu| < 1$.

Proof. Let $(E - P)(X) = \nu X$ with $X \neq 0$. Then

$$E(X) = \nu X + P(X).$$

If $\operatorname{tr}(X) = 0$, then $P(X) = 0$, so $E(X) = \nu X$ and hence ν is an eigenvalue of E with $|\nu| \leq 1$. If $|\nu| = 1$, primitivity forces $\nu = 1$, so X is a trace-zero fixed point of E; by assumption this implies $X = 0$, a contradiction. Thus $|\nu| < 1$.

If instead $\operatorname{tr}(X) \neq 0$, trace preservation and $\operatorname{tr}(P(X)) = \operatorname{tr}(X)$ give

$$\operatorname{tr}(X) = \operatorname{tr}(E(X)) = \nu \operatorname{tr}(X) + \operatorname{tr}(P(X)) = \nu \operatorname{tr}(X) + \operatorname{tr}(X),$$

so $\nu \operatorname{tr}(X) = 0$ and therefore $\nu = 0$. In either case, $|\nu| < 1$. $\square$

Chapter 5

Perron–Frobenius Theory for Channels and Transfer Maps

This chapter establishes the Perron–Frobenius theory used throughout the blueprint: existence, positive definiteness, and uniqueness of fixed points for injective and irreducible transfer maps, canonical gauge constructions, and the general PF eigenvector existence theorem for positive maps. The conceptual source is Wolf's treatment of irreducible positive maps [Wol12, Ch. 6, especially Thms. 6.2–6.5] and [EHK78].

5.1 Positive definiteness

Remark 5.1. Wolf's general theory treats irreducible positive maps first ([Wol12, Thm. 6.2]), then derives non-degeneracy and positive definiteness ([Wol12, Thm. 6.3]). The injective-tensor version (Theorem 5.2) uses a shorter direct kernel argument; the irreducible CP-map version (Theorem 5.3) follows via the support-projection route of [Wol12, Thm. 6.2].

Theorem 5.2 (PSD fixed point is positive definite)**.** *Let A be an injective MPS tensor and let $\rho \geq 0$, $\rho \neq 0$, be a fixed point of the transfer map $\mathcal{E}_A(\rho) = \rho$. Then ρ is positive definite.*

Proof. Suppose $v^\dagger \rho v = 0$ for some $v \neq 0$. Since $\rho \geq 0$, this implies $\rho v = 0$. The fixed-point equation gives $v^\dagger \rho v = \sum_i ((A^i)^\dagger v)^\dagger \rho ((A^i)^\dagger v)$. Each term is nonneg; if the sum is zero, then $\rho (A^i)^\dagger v = 0$ for all i: the kernel of ρ is invariant under all $(A^i)^\dagger$. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, we have $\rho M v = 0$ for every matrix M. Given any vector w, choose M with $Mv = w$. Then $\rho w = 0$ for every w, so $\rho = 0$, contradicting $\rho \neq 0$. In [Wol12, Thm. 6.3, item 2], the same conclusion is reached via the growth condition $(\mathbb{1} + T)^{d-1}(X) > 0$ of [Wol12, Thm. 6.2, condition (2)]; here the direct kernel argument is shorter under injectivity. □

Theorem 5.3 (PSD fixed point is PD — irreducible version)**.** *If $\mathcal{E}_A$ is irreducible and $\rho \geq 0$, $\rho \neq 0$ is a fixed point, then ρ is positive definite.*

Proof. Suppose ρ is not positive definite. Then ρ has a nontrivial kernel; let Q be the orthogonal projection onto the support of ρ (the complement of the kernel). A direct computation using the fixed-point equation $\mathcal{E}_A(\rho) = \rho$ shows that Q is an invariant projection for $\mathcal{E}_A$: it satisfies $Q\,\mathcal{E}_A(QXQ)\,Q = \mathcal{E}_A(QXQ)$ for all X. By irreducibility of $\mathcal{E}_A$ (Definition 4.23), every invariant projection is 0 or $\mathbb{1}$. If $Q = 0$ then $\rho = 0$, contradicting $\rho \neq 0$. If $Q = \mathbb{1}$ then ρ is positive definite, contradicting our assumption. Hence ρ must be positive definite. This is the support-projection direction of [Wol12, Thm. 6.2, condition (1)]. $\square$

Theorem 5.4 (Orthogonal trace condition)**.** *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, and let $A, B \geq 0$ be nonzero positive semidefinite matrices with $\mathrm{tr}(BA) = 0$. Then there exists t with $1 \leq t \leq D-1$ such that*

$$\mathrm{tr}(B \cdot E^t(A)) > 0.$$

Proof. The growth theorem for irreducible CP maps ([Wol12, Thm. 6.2, item 2]) gives $(\mathbb{1} + E)^{D-1}(A) > 0$ (positive definite). Since $B \geq 0$ is nonzero, $\mathrm{tr}(B \cdot (\mathbb{1}+E)^{D-1}(A)) > 0$. Expanding by the binomial theorem:

$$\mathrm{tr}\Big(B \cdot \sum_{k=0}^{D-1} \tbinom{D-1}{k} E^k(A)\Big) > 0.$$

Each term $\mathrm{tr}(B \cdot E^k(A)) \geq 0$ since both factors are PSD. The $k = 0$ term vanishes by hypothesis ($\mathrm{tr}(BA) = 0$). Hence at least one $k \in \{1, \ldots, D-1\}$ contributes a strictly positive term. This is [Wol12, Thm. 6.2, item 4]: item (2) (growth condition) implies item (4) by a binomial expansion and PSD positivity argument. $\square$

5.2 Uniqueness

Theorem 5.5 (Uniqueness of PSD fixed point)**.** *Let A be an injective MPS tensor. If $\rho, \sigma \geq 0$ are both nonzero fixed points of $\mathcal{E}_A$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. Both ρ and σ are positive definite by Theorem 5.2. Write $\rho = SS^\dagger$ via a spectral square root and set $H = (S^\dagger)^{-1}\sigma S^{-1}$, which is Hermitian and positive definite. Let c_0 be the minimal eigenvalue of H. Then $\tau := \sigma - c_0\rho$ is a PSD fixed point which is *not* positive definite (it has a zero eigenvalue by construction of c_0). By Theorem 5.2, τ must be zero, giving $\sigma = c_0\rho$. **Relation to Wolf:** In [Wol12, Thm. 6.3, item 2], uniqueness is proved via the growth condition of [Wol12, Thm. 6.2]. The same minimal-eigenvalue argument applies here, using the injective-tensor positive-definiteness (Theorem 5.2) in place of the growth condition. $\square$

Remark 5.6. The proof produces a positive real scalar $c_0 > 0$: it is the minimal eigenvalue of the positive definite Hermitian matrix $H = (S^\dagger)^{-1}\sigma S^{-1}$.

Theorem 5.7 (Uniqueness of positive eigenvalue for irreducible CP maps)**.** *Let $D \geq 1$ and let E be an irreducible completely positive map on $M_D(\mathbb{C})$. Suppose $\rho, \sigma \geq 0$ are nonzero positive semidefinite matrices satisfying $E(\rho) = r_1\rho$ and $E(\sigma) = r_2\sigma$ for real scalars $r_1, r_2 > 0$. Then $r_1 = r_2$.*

Proof. Extract Kraus operators K for E. Irreducibility of E passes to the adjoint transfer map $E^\dagger(\,\cdot\,) = \sum_i K_i^\dagger\,(\,\cdot\,)\,K_i$, which is a positive CP map. The Perron–Frobenius existence theorem (Theorem 5.15) together with the irreducible PSD-to-PosDef upgrade (Theorem 5.3) gives a positive definite eigenvector $\tau > 0$ with eigenvalue $t > 0$: $E^\dagger(\tau) = t\,\tau$. For any nonzero PSD X with $E(X) = sX$, the trace pairing identity yields

$$s \cdot \mathrm{tr}(\tau X) = \mathrm{tr}(\tau \cdot E(X)) = \mathrm{tr}(E^\dagger(\tau) \cdot X) = t \cdot \mathrm{tr}(\tau X).$$

Since $\tau > 0$ and $X \geq 0$, $X \neq 0$, the scalar $\mathrm{tr}(\tau X) > 0$, so $s = t$. Applying this to both (ρ, r_1) and (σ, r_2) gives $r_1 = t = r_2$. **Relation to Wolf:** This is [Wol12, Thm. 6.3, item 3], establishing non-degeneracy of the spectral-radius eigenvalue for irreducible CP maps. The proof follows Wolf's dual-map trace argument. $\square$

5.3 Existence and assembly

Theorem 5.8 (Existence of PSD fixed point)**.** *Assume $D \geq 1$. Let A be an MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (so that $\mathcal{E}_A$ is trace-preserving). Then there exists $\rho \geq 0$, $\rho \neq 0$, with $\mathcal{E}_A(\rho) = \rho$.*

Proof. Under the normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, the transfer map is a quantum channel (Theorem 4.26). The Cesàro fixed-point theorem (Theorem 4.30) then provides the fixed point; injectivity is not needed for this step. □

Theorem 5.9 (Quantum Perron–Frobenius)**.** *Assume $D \geq 1$. Let A be an injective MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then the transfer map $\mathcal{E}_A$ has a unique PSD fixed point ρ (up to scaling), and ρ is positive definite.*

Proof. Combine Theorems 5.8, 5.2, and 5.5. □

5.4 Right- and left-canonical gauges

Theorem 5.10 (Right-canonical (unital) gauge)**.** *Let $\mathcal{E}_A(\rho) = \rho$ and let S be an invertible matrix with $SS^\dagger = \rho$. Define the gauged operators $A'^i = S^{-1} A^i S$. Then $\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$, i.e. the gauged Kraus map is unital. This is the right-canonical normalization used later in the blueprint.*

Proof. Expand each summand: $(S^{-1}A^iS)(S^{-1}A^iS)^\dagger = S^{-1}A^i \underbrace{SS^\dagger}_{\rho} (A^i)^\dagger (S^\dagger)^{-1}$. Summing over i and using $\sum_i A^i \rho (A^i)^\dagger = \rho$ gives $S^{-1}\rho(S^\dagger)^{-1} = S^{-1}SS^\dagger(S^\dagger)^{-1} = \mathbb{1}$. □

Theorem 5.11 (Left-canonical (trace-preserving) gauge)**.** *Let σ be a fixed point of the adjoint transfer map $\sum_i (A^i)^\dagger \sigma A^i = \sigma$, and let S be invertible with $S^\dagger S = \sigma$. Define $A'^i = SA^iS^{-1}$. Then $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$, i.e. the gauged Kraus map is trace-preserving. This is the left-canonical normalization used later in the blueprint.*

Proof. Expand each summand: $(SA^iS^{-1})^\dagger(SA^iS^{-1}) = (S^\dagger)^{-1}(A^i)^\dagger \underbrace{S^\dagger S}_{\sigma} A^iS^{-1}$. Summing over i and using $\sum_i (A^i)^\dagger \sigma A^i = \sigma$ gives $(S^\dagger)^{-1}\sigma S^{-1} = (S^\dagger)^{-1}S^\dagger SS^{-1} = \mathbb{1}$. □

Remark 5.12. The right-canonical gauge of Theorem 5.10 and the left-canonical gauge of Theorem 5.11 are generally different similarity transforms: one is built from a fixed point of $\mathcal{E}_A$, the other from a fixed point of the adjoint transfer map. This section does not claim that a single gauge makes the transfer map simultaneously unital and trace-preserving.

5.5 Similarity preserves irreducibility

Lemma 5.13 (Similarity transform preserves irreducibility)**.** *Let E be an irreducible completely positive map on $M_D(\mathbb{C})$, let $C \in M_D(\mathbb{C})$ be invertible ($\det C \neq 0$), and let $c > 0$. Define the similarity-transformed map*

$$E'(X) \;=\; c \cdot C^{-1}\, E(C\, X\, C^\dagger)\, (C^\dagger)^{-1}.$$

Then E' is also irreducible.

Proof. Suppose Q is an invariant projection for E': $Q \neq 0$, $Q \neq \mathbb{1}$, and $(\mathbb{1} - Q)E'(QXQ)Q = 0$ for all X. Setting $R = CQC^\dagger$, the support projection P of R is an invariant projection for E. Irreducibility of E forces $P = 0$ or $P = \mathbb{1}$. The invertibility of C then forces $Q = 0$ or $Q = \mathbb{1}$, a contradiction. This is the full similarity version of [Wol12, Prop. 6.6]; the scalar case ($C = \mathbb{1}$) gives the scaling result stated next. □

Theorem 5.14 (Scaling preserves irreducibility)**.** *If E is an irreducible map and $c \neq 0$, then cE is also irreducible. This is the scalar special case ($C = \mathbb{1}$) of [Wol12, Prop. 6.6].*

Proof. An invariant projection for cE is an invariant projection for E (scale out the nonzero constant from the commutation relation). □

5.6 Perron–Frobenius eigenvector existence

This section establishes the existence of a positive definite eigenvector for the adjoint transfer map of an irreducible MPS tensor, and uses it to construct a TP-gauge normalization. The PSD-eigenvector step corresponds to [Wol12, Thm. 6.5] (general positive maps) and [EHK78]; the upgrade to positive definiteness uses the irreducible-map theory of [Wol12, Thm. 6.3]. The TP-gauge application follows [CPGSV17, Appendix A].

Theorem 5.15 (PSD eigenvector for positive maps)**.** *Let $E : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ be a positive linear map with $D > 0$, and assume that $E(\sigma) \neq 0$ for every nonzero PSD matrix σ. Then there exist a nonzero PSD matrix ρ and a positive real $r > 0$ such that $E(\rho) = r\rho$.*

This is the finite-dimensional Perron–Frobenius theorem for positive maps on matrix algebras ([Wol12, Thm. 6.5]; see also [EHK78]). The proof applies Brouwer's fixed-point theorem to the normalization map $\rho \mapsto E(\rho)/\operatorname{tr}(E(\rho))$ on the compact convex set of density matrices.

Theorem 5.16 (Adjoint transfer map is nonzero)**.** *If some Kraus operator $A^i \neq 0$, then the adjoint transfer map $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is nonzero.*

Proof. If $\mathcal{E}_A^\dagger = 0$ then $\mathcal{E}_A^\dagger(\mathbb{1}) = 0$, so $\sum_i (A^i)^\dagger A^i = 0$. Since each summand $(A^i)^\dagger A^i$ is PSD, each $(A^i)^\dagger = 0$ and hence each $A^i = 0$. □

Theorem 5.17 (PosDef adjoint eigenvector for irreducible tensors)**.** *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive definite matrix σ and a positive real $r > 0$ such that $\mathcal{E}_A^\dagger(\sigma) = r\sigma$.*

This combines the general PF existence ([Wol12, Thm. 6.5]) with the irreducible upgrade to positive definiteness ([Wol12, Thm. 6.3, item 2]); the application to the adjoint transfer map follows [CPGSV17, Appendix A].

Proof. By Theorem 5.16, $\mathcal{E}_A^\dagger \neq 0$. Since $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$ is a Kraus map, it is positive. Irreducibility of A transfers to the adjoint transfer map, and together with $\mathcal{E}_A^\dagger \neq 0$ this implies $\mathcal{E}_A^\dagger(\tau) \neq 0$ for every nonzero PSD matrix τ. Hence Theorem 5.15 applies to $\mathcal{E}_A^\dagger$, giving $\sigma_0 \geq 0$ (PSD, nonzero) and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma_0) = r\sigma_0$. Rescale: set $T^i = r^{-1/2}(A^i)^\dagger$. Then $\mathcal{E}_T(\sigma_0) = \sigma_0$, and $\mathcal{E}_T$ is irreducible (by Theorem 5.14, since irreducibility of $\mathcal{E}_A$ transfers to the adjoint and is preserved under scaling). Since $\mathcal{E}_T$ is irreducible with a PSD fixed point, the PSD-to-PosDef upgrade (Theorem 5.3) gives σ_0 positive definite. □

Theorem 5.18 (TP-gauge existence for irreducible tensors)**.** *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real r, a positive definite matrix σ, and a gauge-equivalent tensor B such that*

$$B^i = \sigma^{1/2} \cdot (r^{-1/2} A^i) \cdot \sigma^{-1/2}$$

is trace-preserving: $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

This is the "spectral rescaling + TP gauge" step of [CPGSV17, Appendix A]. The similarity is generally non-unitary.

Proof. By Theorem 5.17, obtain σ positive definite and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma) = r\sigma$. Set $c = r^{-1/2}$ and $A'^i = cA^i$. Then $\mathcal{E}_{A'}^\dagger(\sigma) = \sigma$. By the TP gauge construction (Theorem 5.11), $B^i = \sigma^{1/2} A'^i \sigma^{-1/2}$ satisfies $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. $\square$

Chapter 6

Spectral Gap and Block Separation

This chapter proves that the MPV overlaps between distinct (non-gauge-phase-equivalent) MPS blocks decay to zero as system size grows. It also proves the corresponding self-overlap convergence to 1 under the complementary spectral-gap hypothesis that encodes primitivity in the cases of interest. The argument is based on a spectral analysis of the *mixed transfer operator*, following [PGVWC07].

The algebraic mixed-transfer identities in the first two sections do not use normalization. Starting with the spectral-radius estimates, we assume the trace-preserving normalization

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

6.1 Mixed transfer operator

Definition 6.1 (Mixed transfer operator)**.** The *mixed transfer operator* (or *cross transfer operator*) for two MPS tensors A and B of the same bond dimension D is the linear map $F_{AB} : M_D(\mathbb{C}) \to M_D(\mathbb{C})$ defined by

$$F_{AB}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Definition 6.2 (Rectangular mixed transfer operator)**.** For MPS tensors A of bond dimension D_1 and B of bond dimension D_2 (possibly $D_1 \neq D_2$), the *rectangular mixed transfer operator* is $F^{\mathrm{rect}}_{AB} : M_{D_1 \times D_2}(\mathbb{C}) \to M_{D_1 \times D_2}(\mathbb{C})$ defined by

$$F^{\mathrm{rect}}_{AB}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Remark 6.3 (Self-transfer). Setting $B = A$ recovers the ordinary transfer map: $F_{AA} = \mathcal{E}_A$.

Theorem 6.4 (Iterated mixed transfer)**.** *For all $X \in M_D(\mathbb{C})$ and $N \geq 0$,*

$$F^N_{AB}(X) = \sum_{\sigma \in \{0,\ldots,d-1\}^N} A^\sigma X (B^\sigma)^\dagger,$$

where $A^\sigma = A^{\sigma_1} \cdots A^{\sigma_N}$ denotes the word evaluation (Definition 2.2).

Proof. Induction on N: the base case is $F^0(X) = X$; the inductive step applies F_{AB} to each summand and re-indexes via $\{0, \dots, d-1\}^{N+1} \cong \{0, \dots, d-1\} \times \{0, \dots, d-1\}^N$. $\square$

Theorem 6.5 (Matrix trace of iterated mixed transfer at the identity)**.** *For $N \geq 0$,*

$$\mathrm{tr}(F_{AB}^N(\mathbb{1})) = \sum_\sigma \mathrm{tr}(A^\sigma (B^\sigma)^\dagger).$$

This is the matrix trace of F_{AB}^N evaluated at the identity matrix. It should not *be confused with the operator trace $\mathrm{Tr}(F_{AB}^N)$, which equals the MPV overlap (Theorem 6.7).*

Proof. Apply the matrix trace to each summand in Theorem 6.4 with $X = \mathbb{1}$: $\mathrm{tr}(F_{AB}^N(\mathbb{1})) = \sum_\sigma \mathrm{tr}(A^\sigma \cdot \mathbb{1} \cdot (B^\sigma)^\dagger) = \sum_\sigma \mathrm{tr}(A^\sigma (B^\sigma)^\dagger)$. $\square$

6.2 MPV overlap as transfer trace

Recall from Definition 2.31 that

$$O_{AB}(N) = \sum_{\sigma \in \{0,\dots,d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

This section rewrites that scalar as the operator trace of the mixed transfer power.

Lemma 6.6 (Trace expansion over matrix units)**.** *For any linear endomorphism T on $M_D(\mathbb{C})$, the operator trace can be expanded as*

$$\mathrm{Tr}(T) = \sum_{p,q=0}^{D-1} (T(E_{pq}))_{pq},$$

where E_{pq} is the matrix with 1 in position (p, q) and 0 elsewhere.

Proof. The operator trace $\mathrm{Tr}(T)$ equals the sum of $\mathrm{tr}(T(E_{pq}) \cdot E_{qp})$ over all (p, q), and $\mathrm{tr}(M \cdot E_{qp}) = M_{pq}$. $\square$

Theorem 6.7 (Overlap equals transfer trace)**.** *For tensors A, B of bond dimension $D \geq 1$,*

$$O_{AB}(N) = \mathrm{Tr}(F_{AB}^N).$$

Proof. Expand the operator trace using Lemma 6.6 as a sum over matrix units E_{pq}. Apply Theorem 6.4 to evaluate $F_{AB}^N(E_{pq})$, extract the (p, q)-entry, and reassemble. The resulting double sum over (p, q) and σ equals $\sum_\sigma V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = O_{AB}(N)$. $\square$

Theorem 6.8 (Rectangular overlap as transfer trace)**.** *For tensors A of bond dimension $D_1 \geq 1$ and B of bond dimension $D_2 \geq 1$,*

$$O_{AB}(N) = \mathrm{Tr}((F_{AB}^{\mathrm{rect}})^N).$$

Proof. Expand the operator trace over the rectangular matrix-unit basis. Then apply the rectangular iterated-transfer formula and sum over the matrix-unit indices exactly as in Theorem 6.7. $\square$

Lemma 6.9 (Word trace as entrywise inner product)**.** *For a word $\sigma \in \{0, \dots, d-1\}^N$,*

$$\mathrm{tr}(A^\sigma (B^\sigma)^\dagger) = \sum_{j,k=0}^{D-1} (A^\sigma)_{jk} \, \overline{(B^\sigma)_{jk}}.$$

Proof. Expand the matrix trace and the (j, j)-entry of $A^\sigma (B^\sigma)^\dagger$. $\square$

6.3 Eigenvalue bound and spectral gap

The eigenvalue bound comes from a uniform Frobenius-norm estimate obtained directly from the trace-preserving normalization. For the rigidity theorem, one gauges A and B separately using positive fixed points of their transfer maps to obtain unital Kraus families, and then applies the Kadison–Schwarz equality argument to a block embedding of a mixed-transfer eigenvector.

Theorem 6.10 (Eigenvalue bound)**.** *Let A, B be normalized MPS tensors. Every eigenvalue μ of F_{AB} satisfies $|\mu| \leq 1$.*

Proof. Expand $F^n_{AB}(X)$ as a sum over words and use Cauchy–Schwarz, together with $\sum_i (A^i)^\dagger A^i = \sum_i (B^i)^\dagger B^i = \mathbb{1}$, to obtain a uniform Frobenius-norm bound $\|F^n_{AB}(X)\|_{\mathrm{HS}} \leq C_D \|X\|_{\mathrm{HS}}$ for all n, where C_D depends only on D. If $F_{AB}(X) = \mu X$ and $X \neq 0$, then $|\mu|^n \|X\|_{\mathrm{HS}} = \|F^n_{AB}(X)\|_{\mathrm{HS}} \leq C_D \|X\|_{\mathrm{HS}}$ for every n. Hence $|\mu| \leq 1$.

The argument uses only the trace-preserving normalization and does not appeal to Perron–Frobenius theory. □

Theorem 6.11 (Spectral radius bound)**.** $\rho(F_{AB}) \leq 1$ *for normalized tensors.*

Proof. The spectral radius is the supremum of $|\mu|$ over all eigenvalues, and each satisfies $|\mu| \leq 1$ by Theorem 6.10. □

Theorem 6.12 (Spectral radius ≥ 1 implies gauge-phase equivalence)**.** *Let A, B be injective normalized MPS tensors. If $\rho(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.*

Proof. Since $\rho(F_{AB}) \leq 1$ (Theorem 6.11) and $\rho(F_{AB}) \geq 1$ by hypothesis, there exists an eigenvalue μ with $|\mu| = 1$ and eigenvector $X \neq 0$: $F_{AB}(X) = \mu X$.

Choose positive fixed points of the transfer maps of A and B (Theorem 5.9) and gauge the two tensors separately to unital Kraus families (Theorem 5.10). Gauge X accordingly and embed it as an off-diagonal block matrix for the block Kraus family with diagonal blocks A'^i and B'^i. The KS equality for peripheral eigenvectors (Theorem 4.21) and the resulting Kraus commutation relation (Theorem 4.20) yield intertwining identities between A'^i, B'^i, and the gauged eigenvector. Injectivity and the multiplicative-domain argument force this intertwiner to be invertible; Skolem–Noether (Theorem 3.9) then yields $B^i = \bar{\mu}\, X\, A^i\, X^{-1}$ for all i, establishing gauge-phase equivalence (Definition 2.13).

The crucial point is that one gauges the individual transfer maps to unital form; one does not require the mixed transfer map itself to be simultaneously unital and trace-preserving. □

Theorem 6.13 (Strict spectral gap)**.** *Let A, B be injective normalized MPS tensors that are* not *gauge-phase equivalent. Then $\rho(F_{AB}) < 1$.*

Proof. $\rho(F_{AB}) \leq 1$ by Theorem 6.11. If $\rho(F_{AB}) = 1$, Theorem 6.12 gives gauge-phase equivalence, contradicting the hypothesis. □

6.4 Spectral gap — rectangular case

Theorem 6.14 (Rectangular spectral gap)**.** *Let A be an injective normalized tensor of bond dimension D_1 and B an injective normalized tensor of bond dimension D_2, with $D_1 \neq D_2$. Then* $\rho(F^{\mathrm{rect}}_{AB}) < 1$.

Proof. The same word-expansion argument as in Theorem 6.10 gives $|\mu| \leq 1$ for every eigenvalue of F_{AB}^{rect}. If $\rho(F_{AB}^{\text{rect}}) = 1$, choose a modulus-one eigenvector X.

Gauge A and B separately to unital Kraus families and apply the same block-embedding Kadison–Schwarz argument as in the square case. It shows that $\ker(X)$ is invariant under all matrices on $\mathbb{C}^{D_2}$, so X is injective and therefore $D_2 \leq D_1$. Applying the same argument to $X^\dagger$ gives $D_1 \leq D_2$. Hence $D_1 = D_2$, contradicting the hypothesis. ☐

Theorem 6.15 (Rectangular overlap decay). *If $D_1 \neq D_2$ and both tensors are injective and normalized, then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. The spectral gap $\rho(F_{AB}^{\text{rect}}) < 1$ (Theorem 6.14) implies $(F_{AB}^{\text{rect}})^N \to 0$, so the operator trace converges to 0. By Theorem 6.8, $O_{AB}(N) = \text{Tr}((F_{AB}^{\text{rect}})^N) \to 0$. ☐

6.5 MPV overlap decay

Theorem 6.16 (Transfer powers converge to zero). *Let A, B be injective normalized tensors that are not gauge-phase equivalent. Then $F_{AB}^n(X) \to 0$ for every $X \in M_D(\mathbb{C})$.*

Proof. The spectral gap $\rho(F_{AB}) < 1$ (Theorem 6.13) implies $\|F_{AB}^n\|_{\text{op}} \to 0$ by the Gelfand formula, so $F_{AB}^n(X) \to 0$ for every X. ☐

Theorem 6.17 (Overlap decay). *Let A, B be injective normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \to 0$ as $N \to \infty$.*

Proof. By Theorem 6.7, $O_{AB}(N) = \text{Tr}(F_{AB}^N)$. Expand the operator trace over matrix units (Lemma 6.6): $\text{Tr}(F_{AB}^N) = \sum_{p,q} (F_{AB}^N(E_{pq}))_{pq}$. By Theorem 6.16, each $F_{AB}^N(E_{pq}) \to 0$ entry-wise. Since the sum is finite, $O_{AB}(N) \to 0$. ☐

6.6 Block separation

These asymptotic facts are the ingredients used later in the canonical-form separation arguments.

Remark 6.18 (Cross-correlation decay). If A and B are injective normalized tensors that are not gauge-phase equivalent, then for every $X \in M_D(\mathbb{C})$ one has $\text{tr}(F_{AB}^N(X)) \to 0$ as $N \to \infty$. This is the scalar trace consequence of the operator convergence $F_{AB}^N(X) \to 0$ from Theorem 6.16.

Remark 6.19 (Self-correlation persists). If ρ is a fixed point of $\mathcal{E}_A$, then $\text{tr}(\mathcal{E}_A^N(\rho)) = \text{tr}(\rho)$ for all $N \geq 0$. In particular this applies to the distinguished QPF fixed point from Theorem 5.9.

6.7 Primitive overlap convergence (spectral-gap form)

The theorems in this section are stated directly in terms of the complementary map $E - P$. This is the analytic form needed for the later overlap argument. For primitive normalized transfer maps, Chapter 4 supplies this spectral-gap input under explicit hypotheses.

Theorem 6.20 (Complementary spectral gap implies trace convergence to 1). *Let E be trace-preserving with fixed point ρ ($\text{tr}(\rho) \neq 0$) and let P be the fixed-point projection (Definition 4.31). If $\rho_{\text{spec}}(E - P) < 1$, then $\text{Tr}(E^n) \to 1$ as $n \to \infty$.*

Proof. By the power decomposition (Theorem 4.32), $E^n = P + (E - P)^n$ for $n \geq 1$. Since $\rho_{\text{spec}}(E - P) < 1$, the Gelfand formula gives $(E - P)^n \to 0$, so $\text{Tr}(E^n) \to \text{Tr}(P)$. The operator trace of P equals 1: since $P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)}\rho$, we have $\text{Tr}(P) = \frac{\text{tr}(\rho)}{\text{tr}(\rho)} = 1$. ☐

Theorem 6.21 (Self-overlap convergence from a complementary spectral gap)**.** *Let A be a normalized MPS tensor with a nonzero PSD fixed point ρ of the transfer map. If $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$ (where P is the fixed-point projection), then $O_{AA}(N) \to 1$ as $N \to \infty$.*

Proof. By Theorem 6.7 (with $A = B$), $O_{AA}(N) = \mathrm{Tr}(\mathcal{E}_A^N)$. By Theorem 6.20, $\mathrm{Tr}(\mathcal{E}_A^N) \to 1$. $\square$

Chapter 7

Wielandt Bound

This chapter proves a quantum analog of Wielandt's inequality: for a normal MPS tensor with bond dimension D, the cumulative span of matrix products stabilises after at most D^2 steps. The results follow [SPGWC10].

7.1 Cumulative span

Definition 7.1 (Word span)**.** The *word span* at length n is the subspace

$$S_n(A) = \operatorname{span}_{\mathbb{C}}\{A^w : |w| = n\} \subseteq M_D(\mathbb{C}).$$

This is [SPGWC10, Eq. (1)].

Definition 7.2 (Cumulative span)**.** The *cumulative span* at level n is

$$T_n(A) = S_0(A) + S_1(A) + \cdots + S_n(A).$$

Lemma 7.3 (Monotonicity of cumulative span)**.** *$T_n(A) \leq T_{n+1}(A)$ for all n.*

Proof. T_{n+1} includes all words of length $\leq n+1$, which contains all words of length $\leq n$. ☐

Lemma 7.4 (Dimension bound)**.** *$\dim T_n(A) \leq D^2$ for all n.*

Proof. $T_n(A) \subseteq M_D(\mathbb{C})$ and $\dim M_D(\mathbb{C}) = D^2$. ☐

Theorem 7.5 (Cumulative span stabilisation)**.** *If $T_n(A) = T_{n+1}(A)$, then $T_m(A) = T_n(A)$ for all $m \geq n$.*

Proof. If $T_n = T_{n+1}$, then $S_{n+1} \subseteq T_n$. Left-multiplying any element of S_{n+1} by A^i gives an element of S_{n+2}, so $S_{n+2} \subseteq T_{n+1} = T_n$. By induction, $S_m \subseteq T_n$ for all $m > n$, hence $T_m = T_n$. ☐

Theorem 7.6 (Dimension growth)**.** *If $T_n(A) \neq T_{n+1}(A)$, then $\dim T_{n+1}(A) > \dim T_n(A)$.*

Proof. $T_n \leq T_{n+1}$ (Lemma 7.3) and $T_n \neq T_{n+1}$ imply strict submodule inclusion, hence strict rank increase. ☐

Lemma 7.7 (Word span characterises block injectivity)**.** *$S_L(A) = M_D(\mathbb{C})$ if and only if A is L-block injective.*

Proof. Both conditions assert that the span of all products of length L equals $M_D(\mathbb{C})$. ☐

7.2 Fitting decomposition

Definition 7.8 (Fitting decomposition)**.** A *Fitting decomposition* of a linear endomorphism $f : V \to V$ on a finite-dimensional vector space over an algebraically closed field consists of:

1. f is nilpotent on the generalized 0-eigenspace,
2. f is invertible on each generalized μ-eigenspace for $\mu \neq 0$,
3. the generalized eigenspaces span V,
4. the generalized eigenspaces are linearly independent.

Theorem 7.9 (Fitting decomposition exists)**.** *Every linear endomorphism on a finite-dimensional vector space over an algebraically closed field admits a Fitting decomposition.*

Proof. Nilpotency on the zero generalized eigenspace follows from the definition of generalized eigenspaces. Invertibility on nonzero generalized eigenspaces follows because $f - \mu$ is nilpotent there, so $f = \mu(1 - (1 - f/\mu))$ is invertible. Spanning and independence are standard results for generalized eigenspaces over algebraically closed fields. In [SPGWC10] and [Wol12, Lemma 6.3(b)], the Jordan normal form is used directly. The argument is phrased in terms of generalized eigenspaces instead. □

Theorem 7.10 (Nilpotent power bound)**.** *If f is nilpotent on a space of dimension n, then $f^n = 0$.*

Proof. A nilpotent endomorphism on an n-dimensional space has nilpotency index $\leq n$. □

7.3 Nonzero trace product

Theorem 7.11 (Cumulative span reaches $M_D(\mathbb{C})$ — explicit bound)**.** *If A is normal, then there exists $n \leq D^2$ with $T_n(A) = M_D(\mathbb{C})$.*

Proof. By Theorem 7.6, if $T_n \neq T_{n+1}$ then $\dim T_{n+1} > \dim T_n$. Since $\dim T_n \leq D^2$ (Lemma 7.4), after at most D^2 steps either $T_n = M_D(\mathbb{C})$ or T_n has stabilised. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality (which requires $S_{L_0} = M_D(\mathbb{C})$ for some L_0, hence $T_{L_0} = M_D(\mathbb{C})$). □

Theorem 7.12 (Cumulative span reaches $M_D(\mathbb{C})$)**.** *If A is normal, then $T_{D^2}(A) = M_D(\mathbb{C})$.*

Proof. Take n from Theorem 7.11; since $n \leq D^2$ and T is monotone, $T_{D^2} = M_D(\mathbb{C})$. □

Theorem 7.13 (Nonzero trace product)**.** *If A is normal, then there exists a word w of length $\leq D^2$ such that $\operatorname{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1].*

Proof. By Theorem 7.12, $\mathbb{1} \in T_{D^2}(A)$, so $\mathbb{1}$ is a linear combination of word products of length $\leq D^2$. At least one has nonzero trace (since $\operatorname{tr}(\mathbb{1}) = D \neq 0$). □

7.4 Eigenvector extraction

Theorem 7.14 (Nonzero trace implies eigenvector)**.** *If $M \in M_D(\mathbb{C})$ has $\operatorname{tr}(M) \neq 0$, then M has a nonzero eigenvalue $\mu \neq 0$ and a corresponding eigenvector $\varphi \neq 0$ with $M\varphi = \mu\varphi$.*

Proof. A matrix with nonzero trace has a nonzero eigenvalue (since trace = sum of eigenvalues counted with multiplicity). The Fitting decomposition (Theorem 7.9) ensures the corresponding generalized eigenspace is nontrivial, yielding an eigenvector. □

Theorem 7.15 (Eigenvalue extraction)**.** *If A is normal, then there exist a word w_0 with $|w_0| \leq D^2$, a nonzero scalar $\mu \neq 0$, and a nonzero vector $\varphi \neq 0$ such that $A^{w_0}\varphi = \mu\varphi$.*

Proof. By Theorem 7.13, there is a word w_0 with $\operatorname{tr}(A^{w_0}) \neq 0$ and $|w_0| \leq D^2$. By Theorem 7.14, A^{w_0} has a nonzero eigenvector. □

7.5 Eigenvector spreading

Definition 7.16 (Cumulative vector span)**.** Given an MPS tensor A and a vector $\varphi \in \mathbb{C}^D$, the *cumulative vector span* is

$$K_n(A, \varphi) = \operatorname{span}_{\mathbb{C}}\{A^w\varphi : |w| \leq n\} \subseteq \mathbb{C}^D.$$

This is [SPGWC10, proof of Lemma 2(a)].

Theorem 7.17 (Vector span stabilisation)**.** *If $K_n(A, \varphi) = K_{n+1}(A, \varphi)$, then $K_m(A, \varphi) = K_n(A, \varphi)$ for all $m \geq n$.*

Proof. Same argument as Theorem 7.5: left-multiplication by A^i cannot escape the stabilised span. □

Lemma 7.18 (Vector span from matrix span)**.** *If $T_N(A) = M_D(\mathbb{C})$ and $\varphi \neq 0$, then $K_N(A, \varphi) = \mathbb{C}^D$.*

Proof. Every matrix $M \in M_D(\mathbb{C})$ is in $T_N(A)$, so in particular every rank-one matrix sending φ to any target vector v is in T_N. Applying such matrices to φ recovers all of $\mathbb{C}^D$. □

Theorem 7.19 (Eigenvector spreading)**.** *Let A be a normal MPS tensor and let $A^{i_0}\varphi = \mu\varphi$ with $\mu \neq 0$ and $\varphi \neq 0$. Then $K_{D-1}(A, \varphi) = \mathbb{C}^D$. This is [SPGWC10, Lemma 2(a)].*

Proof. The vector span K_n is monotone and $\dim K_n \leq D$. If $K_n = K_{n+1}$ for some $n < D - 1$, the span stabilises (Theorem 7.17). But $T_{D^2}(A) = M_D(\mathbb{C})$ by Theorem 7.12, so Lemma 7.18 gives $K_{D^2}(A, \varphi) = \mathbb{C}^D$, contradicting early stabilisation. Hence $\dim K_n$ grows by ≥ 1 at each step, reaching D by step $D - 1$. □

7.6 Assembly

Definition 7.20 (Wielandt analysis)**.** A *Wielandt analysis* for a normal MPS tensor A packages: a word w_0 of length $\leq D^2$, a nonzero eigenvalue μ, and a nonzero eigenvector φ of the matrix A^{w_0}, such that $A^{w_0}\varphi = \mu\varphi$.

Theorem 7.21 (Wielandt analysis exists)**.** *Every normal MPS tensor admits a Wielandt analysis.*

Proof. Immediate from Theorem 7.15. ☐

Theorem 7.22 (Wielandt chain)**.** *Every normal MPS tensor admits a Wielandt analysis consisting of data* (w_0, μ, φ) *with* $K_{D-1}(A, \varphi) = \mathbb{C}^D$.

Proof. Combine Theorems 7.21 and 7.19. ☐

Theorem 7.23 (Wielandt bound)**.** *If* A *is a normal MPS tensor with bond dimension* D*, then* $T_{D^2}(A) = M_D(\mathbb{C})$*. In particular, the cumulative span stabilises at index at most* D^2*. This is [SPGWC10, Theorem 1] (which gives the sharper bound* $D^2 - d + 1$ *in certain cases; here we prove the* D^2 *bound).*

Proof. Immediate from Theorem 7.12. ☐

Remark 7.24 (Explicit blocking-length estimates)**.** Theorem 7.23 gives the D^2 cumulative-span bound. The sharper explicit blocking-length estimates from [SPGWC10] — at most D^4 for injectivity and at most $3D^5$ for the bi-canonical form — are not proved here.

7.7 Lemma 2(b): fixed-length matrix spanning

The preceding sections establish the cumulative-spanning part of the Lemma 2(a) argument from [SPGWC10]: eigenvector spreading gives a cumulative vector span $K_{D-1}(A, \varphi) = \mathbb{C}^D$. The next stage is to pass to a span at one fixed length, namely to find n_0 such that $S_{n_0}(A) = M_D(\mathbb{C})$ (i.e. word products of *exactly* one length span all matrices).

Definition 7.25 (Fixed-length vector span)**.** The *fixed-length vector span* at length n is

$$H_n(A, \varphi) = \operatorname{span}_{\mathbb{C}}\{A^w \varphi : |w| = n\} \subseteq \mathbb{C}^D.$$

This is $S_n(A)|\varphi\rangle$ in the notation of [SPGWC10].

Lemma 7.26 (Word span products)**.** $S_m(A) \cdot S_n(A) \subseteq S_{m+n}(A)$*: the product of word spans at lengths* m *and* n *is contained in the word span at length* $m + n$*.*

Proof. Every product $A^{w_1} A^{w_2}$ with $|w_1| = m$ and $|w_2| = n$ equals $A^{w_1 w_2}$ with $|w_1 w_2| = m + n$. ☐

Lemma 7.27 (Eigenvector padding)**.** *Suppose* $A^{i_0}\varphi = \mu\varphi$ *with* $\mu \neq 0$*. Then for all* n*,*

$$K_n(A, \varphi) = H_n(A, \varphi).$$

In particular, if $K_n(A, \varphi) = \mathbb{C}^D$ *then already* $H_n(A, \varphi) = \mathbb{C}^D$*.*

Proof. Any word w with $|w| \leq n$ can be padded to exact length n by appending $n - |w|$ copies of the index i_0. The eigenvector relation gives $A^{w \cdot i_0^k}\varphi = \mu^k \cdot A^w \varphi$, so the padded vector is a nonzero scalar multiple of the original. Hence every generator of K_n lies in H_n. The reverse inclusion $H_n \leq K_n$ is immediate. ☐

Theorem 7.28 (Vector spanning to matrix spanning (assembly step))**.** *Assume:*

1. $H_n(A, \varphi) = \mathbb{C}^D$ *(length-*n *word products applied to* φ *span all of* $\mathbb{C}^D$*), and*
2. *for each standard basis vector* e_j*, the rank-one operator* $|\varphi\rangle\langle e_j|$ *lies in* $S_m(A)$*.*

Then $S_{n+m}(A) = M_D(\mathbb{C})$.

Proof. Fix a matrix unit E_{ij}. By hypothesis (1), there exists $M_i \in S_n(A)$ with $M_i\varphi = e_i$. By hypothesis (2), $|\varphi\rangle\langle e_j| \in S_m(A)$. Their product satisfies

$$M_i \cdot |\varphi\rangle\langle e_j| = |M_i\varphi\rangle\langle e_j| = |e_i\rangle\langle e_j| = E_{ij},$$

and $E_{ij} \in S_{n+m}(A)$ by Lemma 7.26. Since the matrix units span $M_D(\mathbb{C})$, $S_{n+m}(A) = M_D(\mathbb{C})$. $\square$

Lemma 7.29 (Blocking transfer)**.** *If the blocked tensor* $A^{[L]}$ *satisfies* $S_n(A^{[L]}) = M_D(\mathbb{C})$*, then* $S_{nL}(A) = M_D(\mathbb{C})$*.*

Proof. Each length-n word in the blocked alphabet $\{0, \dots, d^L - 1\}$ decodes to a length-nL word in the original alphabet $\{0, \dots, d-1\}$, so $S_n(A^{[L]}) \subseteq S_{nL}(A)$. $\square$

Remark 7.30 (Relation with Lemma 2(b)). In [SPGWC10], Lemma 2(b) combines a rank-one extraction for fixed-length spans with an assembly argument turning fixed-length vector spanning into fixed-length matrix spanning. Theorem 7.28 is the second step. Theorem 7.35 supplies the rank-one input in blocked form, and Theorem 7.36 then gives a blocked, fixed-length word-span saturation result.

7.8 Blocked tensor and rectangular span

This section extends the Lemma 2(b) infrastructure to the *blocked tensor* $A^{[L]}$ and develops the "rectangular span" used to produce rank-one elements.

Theorem 7.31 (Blocking preserves normality)**.** *If* A *is normal and* $L > 0$*, then the blocked tensor* $A^{[L]}$ *is also normal.*

Proof. If $S_N(A) = M_D(\mathbb{C})$ then $S_{NL}(A) = M_D(\mathbb{C})$ (padding by repeated single-letter words). By Lemma 7.32, $S_{NL}(A) \subseteq S_N(A^{[L]})$, so $S_N(A^{[L]}) = M_D(\mathbb{C})$. $\square$

Lemma 7.32 (Chunking: word span embeds into blocked word span)**.** $S_{nL}(A) \subseteq S_n(A^{[L]})$ *for all* n, L*.*

Proof. Every length-nL word in the original alphabet can be split into n consecutive chunks of length L, each of which is a single blocked Kraus operator. By induction on n, using the word span product lemma (Lemma 7.26). $\square$

Theorem 7.33 (Word eigenvector gives blocked eigenvector)**.** *If* $A^{w_0}\varphi = \mu\varphi$ *for a word* w_0 *of length* L*, then* $(A^{[L]})^{j_0}\varphi = \mu\varphi$*, where* j_0 *is the encoded blocked index corresponding to* w_0*.*

Proof. By definition, $(A^{[L]})^{j_0} = A^{w_0}$. $\square$

Theorem 7.34 (Wielandt blocked assembly)**.** *Suppose* A *is normal and* $L > 0$*, and we have:*

1. *a word* w_0 *of length* L *with eigenvector* $A^{w_0}\varphi = \mu\varphi$ ($\mu \neq 0$, $\varphi \neq 0$)*,*
2. *a word* τ_0 *of length* L *with transpose eigenvector* $(A^{\tau_0})^T\psi = \nu\psi$ ($\nu \neq 0$, $\psi \neq 0$)*,*
3. *a rank-one element* $\varphi\psi^T \in S_m(A^{[L]})$ *for some* m*.*

Then $S_{((D-1)+m+(D-1))\cdot L}(A) = M_D(\mathbb{C})$*.*

Proof. Apply the conditional Lemma 2(b) assembly (Theorem 7.28) to the blocked tensor $A^{[L]}$, which is normal by Theorem 7.31. The eigenvector and transpose eigenvector are transferred to single-index eigenvectors of $A^{[L]}$ by Theorem 7.33. Transfer back to the original tensor via Lemma 7.29. $\square$

Theorem 7.35 (Rank-one extraction for blocked tensor)**.** *Suppose A is normal and $N_0 \geq 1$ with $S_{N_0}(A) = M_D(\mathbb{C})$. Then there exist words σ_0, τ_0 of length N_0, nonzero vectors φ, ψ, nonzero scalars μ, ν, and $m \in \mathbb{N}$ such that:*

1. $A^{\sigma_0}\varphi = \mu\varphi$,
2. $(A^{\tau_0})^T\psi = \nu\psi$,
3. $\varphi\psi^T \in S_m(A^{[N_0]})$.

Proof. Set $B = A^{[N_0]}$. Since $S_{N_0}(A) = M_D(\mathbb{C})$, a nonzero-trace word exists (Theorem 7.13), yielding a column eigenvector φ via Theorem 7.14. A row eigenvector ψ is obtained analogously from the transpose tensor. Set $P = (B^{j_0})^D$ and $Q = (B^{k_0})^D$, where j_0, k_0 encode σ_0, τ_0 in the blocked alphabet. The Fitting decomposition ensures $\varphi \in \mathrm{range}(P)$ and $\psi \in \mathrm{range}(Q^T)$, so $\varphi\psi^T \in \mathrm{range}(\mathrm{mulLeft}(P) \circ \mathrm{mulRight}(Q)) \subseteq S_{2D}(B)$. $\square$

Theorem 7.36 (Fixed-length word-span saturation via blocking)**.** *If A is a normal MPS tensor with bond dimension D, then there exists N such that $S_N(A) = M_D(\mathbb{C})$. This gives the fixed-length spanning conclusion corresponding to [SPGWC10, Lemma 2(b)].*

Proof. By normality, some N_0 has $S_{N_0}(A) = M_D(\mathbb{C})$. If $N_0 = 0$ we are done. Otherwise, Theorem 7.35 produces eigenvectors and a rank-one element $\varphi\psi^T \in S_m(A^{[N_0]})$. Theorem 7.34 then gives $S_{((D-1)+m+(D-1))\cdot N_0}(A) = M_D(\mathbb{C})$. $\square$

Remark 7.37 (Blocking route to fixed-length spanning). Theorem 7.36 is obtained by first extracting a rank-one operator in a blocked tensor (Theorem 7.35) and then applying the fixed-length assembly step (Theorem 7.34). Thus the fixed-length span of A is reached through a blocked auxiliary tensor rather than by a direct unblocked argument.

7.9 Primitive MPS tensors

Definition 7.38 (Primitive MPS tensor)**.** Assume $D > 0$. A tensor A together with a matrix $\rho \in M_D(\mathbb{C})$ is *primitive* if

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}, \qquad \rho \geq 0, \qquad \rho \neq 0, \qquad \mathcal{E}_A(\rho) = \rho,$$

and, if P is the fixed-point projection associated to ρ, then $\rho_{\mathrm{spec}}(\mathcal{E}_A - P) < 1$. When the choice of ρ is irrelevant, we simply say that A is a primitive MPS tensor.

Theorem 7.39 (Primitive overlap convergence)**.** *If A is a primitive MPS tensor, then $O_{AA}(N) \to 1$.*

Proof. Immediate from the self-overlap convergence under a complementary spectral gap (Theorem 6.21). $\square$

7.10 Primitivity and normality

Throughout this section, primitivity means the condition of Definition 7.38: the tensor is in TP gauge, it has a nonzero PSD fixed point, and the complementary map has spectral radius < 1.

Theorem 7.40 (Primitive MPS tensors have nonzero word products)**.** *If A is a primitive MPS tensor, then for every $n \geq 0$ there exists a word w of length* exactly *n with $A^w \neq 0$.*

Proof. The PSD fixed point ρ of $\mathcal{E}_A$ satisfies $\mathcal{E}_A^n(\rho) = \rho \neq 0$. Since $\mathcal{E}_A^n(\rho) = \sum_{|w|=n} A^w \rho (A^w)^\dagger$, at least one summand is nonzero, hence some $A^w \neq 0$. ☐

Theorem 7.41 (Iterated transfer does not vanish)**.** *If A is a primitive MPS tensor, then $\mathcal{E}_A^n \neq 0$ for all $n \geq 0$.*

Proof. By Theorem 7.40, there exists a word w of length n with $A^w \neq 0$, so $\mathcal{E}_A^n(\rho)$ has a nonzero summand. ☐

7.10.1 Primitivity-to-normality bridge

The primitivity condition of Definition 7.38 combines TP normalization, a nonzero PSD fixed point, and a complementary spectral gap. It is connected to normality through the following results.

Theorem 7.42 (Fixed-point uniqueness from spectral gap)**.** *If A is a primitive MPS tensor with PSD fixed point ρ, then every fixed point σ of $\mathcal{E}_A$ is proportional to ρ: $\sigma = \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho$.*

Proof. Set $\sigma' = \sigma - \frac{\mathrm{tr}(\sigma)}{\mathrm{tr}(\rho)}\rho$. Then $\mathrm{tr}(\sigma') = 0$ and $(E - P_\rho)(\sigma') = \sigma'$. If $\sigma' \neq 0$, then 1 is an eigenvalue of $E - P_\rho$, contradicting the spectral gap $\rho_{\mathrm{spec}}(E - P_\rho) < 1$. ☐

Theorem 7.43 (Complement powers tend to zero)**.** *If A is a primitive MPS tensor with PSD fixed point ρ, then $(\mathcal{E}_A - P_\rho)^n \to 0$ in operator norm.*

Proof. The spectral radius of $\mathcal{E}_A - P_\rho$ is strictly less than 1 by the spectral-gap hypothesis. Apply the general result that powers of an operator with spectral radius < 1 tend to zero. ☐

Remark 7.44 (Primitivity: spectral gap vs. positive definiteness)*.* The primitivity condition used in §7.10.1 has three parts: TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, a nonzero PSD fixed point ρ, and the spectral gap $\rho_{\mathrm{spec}}(\mathcal{E}_A - P_\rho) < 1$. The notion of "primitive" in [SPGWC10, Proposition 3] additionally requires ρ to be positive definite (full rank). Without positive definiteness, the spectral-gap condition alone does *not* imply irreducibility or normality. A counterexample is $D = 2$, $\rho = \mathrm{diag}(1, 0)$.

With the additional hypothesis $\rho > 0$, Theorem 7.45 gives irreducibility. Together with the Burnside bridge (Chapter 8, §8.4.1) and Theorem 7.51, one obtains

$$\text{primitive} + \rho \text{ PosDef} \;\Rightarrow\; \text{irreducible} \;\Rightarrow\; \mathrm{algSpan}(A) = M_D(\mathbb{C}) \;\Rightarrow\; \text{normal}.$$

The last step uses the aperiodicity condition $\mathbb{1} \in S_1(A)$ from Remark 7.46.

7.10.2 Primitive implies irreducible

Theorem 7.45 (Primitive with PosDef fixed point implies irreducible tensor)**.** *If A is a primitive MPS tensor with PSD fixed point ρ and ρ is positive definite, then A is an irreducible tensor. This is part of [SPGWC10, Proposition 3] (see also [CPGSV17, §2.3]).*

Proof. By contrapositive. Assume A has an invariant projection P ($P \neq 0$, $P \neq \mathbb{1}$, $(\mathbb{1}-P)A^i P = 0$ for all i). Set $\sigma = P\rho P$.

1. *Invariance of σ under $\mathcal{E}_A^n$:* the projection relation $(\mathbb{1}-P)A^i P = 0$ implies $(\mathbb{1}-P)\cdot\mathcal{E}_A^n(\sigma) = 0$ for all n (by induction on n).

2. *Convergence:* By Theorem 7.43, $\mathcal{E}_A^n(\sigma) \to \frac{\operatorname{tr}(\sigma)}{\operatorname{tr}(\rho)}\rho$. Hence $(\mathbb{1} - P) \cdot \frac{\operatorname{tr}(\sigma)}{\operatorname{tr}(\rho)}\rho = 0$.

3. *Nonzero scalar:* Since $P \neq 0$ and ρ is positive definite, $\operatorname{tr}(P\rho P) > 0$, so $\frac{\operatorname{tr}(\sigma)}{\operatorname{tr}(\rho)} \neq 0$. Therefore $(\mathbb{1} - P)\rho = 0$.

4. *Contradiction:* Since ρ is positive definite, it is a unit in $M_D(\mathbb{C})$. From $(\mathbb{1} - P)\rho = 0$ we get $\mathbb{1} - P = 0$, i.e. $P = \mathbb{1}$, contradicting $P \neq \mathbb{1}$.

□

7.11 Cumulative span to word span

Cumulative spanning ($T_N(A) = M_D(\mathbb{C})$, the union of word spans of all lengths $\leq N$) is weaker than normality ($S_L(A) = M_D(\mathbb{C})$ for a single length L). The gap is closed by an aperiodicity hypothesis.

Remark 7.46 (Aperiodicity condition). We say A satisfies the *aperiodicity condition* if $\mathbb{1} \in S_1(A)$, i.e. the identity matrix lies in the span of the Kraus operators $\{A^i\}$. This is an additional hypothesis. The one-sided TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ does *not* imply $\mathbb{1} \in S_1(A)$; it only gives a quadratic identity among the Kraus operators. For the paper-style primitive condition (peripheral-spectrum primitivity together with a positive-definite fixed point), one expects aperiodicity to follow from spectral analysis of the transfer map. That implication is not used in the present Wielandt bridge, so the condition is kept explicit.

Remark 7.47 (Counterexample: cumulative spanning does not imply normality). In $M_2(\mathbb{C})$, let $A^0 = E_{12}$ and $A^1 = E_{21}$ (the off-diagonal matrix units). Then $\operatorname{algSpan}(A) = M_2(\mathbb{C})$ and $T_2(A) = M_2(\mathbb{C})$, but the word spans alternate: $S_n(A)$ contains only off-diagonal matrices for odd n and only diagonal matrices for even n. Hence $S_n(A) \neq M_2(\mathbb{C})$ for every n, and A is not normal. The aperiodicity condition $\mathbb{1} \in S_1(A)$ fails: $S_1(A) = \operatorname{span}_{\mathbb{C}}\{E_{12}, E_{21}\}$.

Theorem 7.48 (Word span monotonicity from aperiodicity)**.** *If $\mathbb{1} \in S_1(A)$, then $S_n(A) \leq S_{n+1}(A)$ for all n.*

Proof. For any $M \in S_n(A)$, $M = M \cdot \mathbb{1} \in S_n(A) \cdot S_1(A) \subseteq S_{n+1}(A)$ by Lemma 7.26. □

Theorem 7.49 (Cumulative span equals word span under aperiodicity)**.** *If $\mathbb{1} \in S_1(A)$, then $T_n(A) = S_n(A)$ for all n.*

Proof. By Theorem 7.48, the word spans are monotone: $S_0 \leq S_1 \leq \cdots$. The supremum $T_n = S_0 + \cdots + S_n$ therefore equals the largest term S_n. □

Theorem 7.50 (Cumulative spanning plus aperiodicity implies normality)**.** *If* $T_N(A) = M_D(\mathbb{C})$ *and* $\mathbb{1} \in S_1(A)$*, then* A *is normal.*

Proof. By Theorem 7.49, $S_N(A) = T_N(A) = M_D(\mathbb{C})$. □

Theorem 7.51 (Algebra span plus aperiodicity implies normality)**.** *If* $\operatorname{algSpan}(A) = M_D(\mathbb{C})$ *and* $\mathbb{1} \in S_1(A)$*, then* A *is normal.*

Proof. Since $\operatorname{algSpan}(A) = M_D(\mathbb{C})$, there exists N with $T_N(A) = M_D(\mathbb{C})$ (by the ascending chain condition on finite-dimensional subspaces). Apply Theorem 7.50. □

Theorem 7.52 (Irreducible tensor plus aperiodicity implies normality)**.** *If* A *is an irreducible tensor,* $D > 0$*, and* $\mathbb{1} \in S_1(A)$*, then* A *is normal. This assembles the full chain:*

$$\textit{irreducible tensor} \xrightarrow{8.20} \textit{irreducible action} \xrightarrow{\textit{Burnside}} \operatorname{algSpan} = M_D(\mathbb{C}) \xrightarrow{7.51} \textit{normal.}$$

This corresponds to [SPGWC10, Theorem 1 / Proposition 3].

Proof. By Theorem 8.20, A acts irreducibly on $\mathbb{C}^D$. By Theorem 8.21, $\operatorname{algSpan}(A) = M_D(\mathbb{C})$. By Theorem 7.51 with the aperiodicity hypothesis, A is normal. □

Theorem 7.53 (Primitive + positive definite fixed point + aperiodicity imply normality)**.** *If* A *is a primitive MPS tensor with fixed point* ρ*,* ρ *is positive definite, and* $\mathbb{1} \in S_1(A)$*, then* A *is normal.*

Proof. By Theorem 7.45, A is an irreducible tensor. Apply Theorem 7.52. □

Remark 7.54 (The normal-canonical route bypasses the aperiodicity bridge). The aperiodicity results in §7.11 are needed only for the block-injective route through Definition 2.19, where one upgrades cumulative spanning to eventual block injectivity. The normal-canonical route used later in Chapters 8–10 does not pass through this step: it works directly with irreducible TP-normalized blocks whose blocked transfer maps are primitive and, once the corresponding weighted block data are available, packages them as normal canonical form in the sense of [CPGSV17]. There are analogous cumulative-span lemmas for blocked irreducible tensors and eigenvector-spreading arguments, but they are auxiliary here and we do not list them separately.

Chapter 8

Canonical Form Reduction

This chapter reduces a general MPS tensor to block-diagonal data via iterated invariant-subspace decomposition. Two reduction schemes are relevant. The block-injective route aims at injective TP-normalized blocks and the canonical form predicate of Chapter 9. The normal-canonical route packages a weighted family of irreducible TP-normalized blocks with primitive transfer maps and pairwise distinct nonzero weight moduli, in the sense of [CPGSV17]. The Perron–Frobenius eigenvector existence and TP-gauge construction are in Chapter 5; this chapter uses them to produce CFII data and the normal canonical form packaging. The reduction draws on [PGVWC07], [CPGSV21], and [Wol12, Ch. 6].

8.1 Block-diagonal assembly

Definition 8.1 (Block-diagonal tensor from blocks)**.** Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block MPS tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, define the block-diagonal tensor $A^i = \bigoplus_{k=1}^r \mu_k A_k^i$ of total bond dimension $D = \sum_k D_k$.

Theorem 8.2 (Per-block single-block FT)**.** *If all block tensors A_k are injective and the per-block same-MPV condition holds (A_k and B_k generate the same MPV family for each k), then each pair (A_k, B_k) is gauge equivalent.*

Proof. Apply the single-block fundamental theorem (Theorem 3.11) to each block independently. □

Theorem 8.3 (Per-block same MPV implies global same MPV)**.** *If A_k and B_k generate the same MPV family for each block k, then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ also generate the same MPV family.*

Proof. By the MPV decomposition (Theorem 2.24), each MPV is $\sum_k \mu_k^N V^{(N)}(A_k)_\sigma$. Per-block equality of MPV coefficients gives global equality. □

Theorem 8.4 (Global gauge from block gauge)**.** *If each pair (A_k, B_k) is gauge equivalent ($B_k^i = X_k A_k^i X_k^{-1}$), then the block-diagonal tensors are gauge equivalent via the block-diagonal matrix $X = \bigoplus_k X_k$.*

Proof. Block-diagonal conjugation: $X(\bigoplus_k \mu_k A_k^i)X^{-1} = \bigoplus_k \mu_k X_k A_k^i X_k^{-1} = \bigoplus_k \mu_k B_k^i$. □

8.2 Transfer map normalization

Theorem 8.5 (Scaling preserves injectivity). *If A is injective and $\zeta \neq 0$, then ζA is injective.*

Proof. Scaling by a nonzero scalar preserves linear independence and hence spanning. □

Theorem 8.6 (Transfer map under scaling). $\mathcal{E}_{\zeta A}(X) = |\zeta|^2 \mathcal{E}_A(X)$.

Proof. Each summand: $(\zeta A^i)X(\zeta A^i)^\dagger = |\zeta|^2 A^i X (A^i)^\dagger$. □

Theorem 8.7 (Phase-scaling preserves normalization). *If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $|\zeta| = 1$, then $\sum_i (\zeta A^i)^\dagger (\zeta A^i) = \mathbb{1}$.*

Proof. $(\zeta A^i)^\dagger(\zeta A^i) = \bar{\zeta}\zeta (A^i)^\dagger A^i = (A^i)^\dagger A^i$, since $\bar{\zeta}\zeta = |\zeta|^2 = 1$. □

Theorem 8.8 (MPV under block normalization). *For any nonzero scaling factors μ_k, write $\mu_k = |\mu_k| \cdot (\mu_k/|\mu_k|)$. The MPVs of $\bigoplus_k \mu_k A_k$ and $\bigoplus_k |\mu_k| \cdot (\mu_k/|\mu_k|)A_k$ agree.*

Proof. By the MPV decomposition (Theorem 2.24), the k-th term is $\mu_k^N V^{(N)}(A_k)_\sigma$. The scaling identity (Lemma 2.12) absorbs the phase into the word evaluation. □

8.3 Invariant subspace decomposition

Theorem 8.9 (Unitary conjugation preserves the MPV family). *If U is a unitary matrix, then A and $U^\dagger A U$ generate the same MPV family.*

Proof. By cyclicity of the trace: $\mathrm{tr}((U^\dagger A^{i_1} U) \cdots (U^\dagger A^{i_N} U)) = \mathrm{tr}(A^{i_1} \cdots A^{i_N})$. □

Definition 8.10 (Support projection). For a positive semi-definite matrix $\rho \geq 0$, the *support projection* P is the orthogonal projection onto the range of ρ. It is constructed via the spectral decomposition: if $\rho = U \operatorname{diag}(\lambda_1, \dots, \lambda_D) U^\dagger$, then $P = U \operatorname{diag}(\mathbf{1}_{\lambda_1 > 0}, \dots, \mathbf{1}_{\lambda_D > 0}) U^\dagger$, where $\mathbf{1}_{\lambda_j > 0}$ is 1 if $\lambda_j > 0$ and 0 otherwise.

Definition 8.11 (Invariant projection predicate). An MPS tensor A *has an invariant projection* if there exists an orthogonal projection P with $P \neq 0$, $P \neq \mathbb{1}$, and $(\mathbb{1} - P)A^i P = 0$ for all i. The negation of this predicate defines an *irreducible tensor* (Definition 8.14).

Theorem 8.12 (PSD fixed point gives invariant projection). *Let $\rho \geq 0$ be a fixed point of $\mathcal{E}_A$, and let P be the support projection of ρ. Then for all i, $(\mathbb{1} - P)A^i P = 0$: the range of P is invariant under all A^i.*

Proof. From $\mathcal{E}_A(\rho) = \rho$ and positivity of each summand $A^i \rho (A^i)^\dagger$, $(\mathbb{1} - P)A^i \rho (A^i)^\dagger (\mathbb{1} - P) = 0$ for each i. Since ρ is supported on the range of P, this forces $(\mathbb{1} - P)A^i P = 0$.

This is the finite-dimensional version of [Wol12, Prop. 6.10] ("Stationary subspaces": the support projection of any stationary state is sub-harmonic, i.e. invariant under the channel). The equivalence between irreducibility and absence of nontrivial invariant projections is [Wol12, Thm. 6.2]; see also [CPGSV21, §IV.A.1]. □

Theorem 8.13 (Two-block decomposition). *If the transfer map has a PSD fixed point ρ that is* not *positive definite (i.e. ρ has a nontrivial kernel), then A is unitarily equivalent to a two-block upper-triangular tensor. Moreover, the diagonal blocks generate the same MPV as A.*

Proof. The support projection P of ρ is neither 0 nor $\mathbb{1}$ (since $\rho \neq 0$ and ρ is not PD). Theorem 8.12 gives $(\mathbb{1} - P)A^i P = 0$ for all i. A unitary change of basis that block-diagonalizes P puts A^i in upper-triangular block form (Theorem 8.9 ensures the MPV is preserved by the conjugation). The trace of the upper-triangular product equals the sum of the block-diagonal traces, so the MPV decomposes into block contributions. □

8.4 Iterated reduction

Definition 8.14 (Irreducible tensor)**.** An MPS tensor A is *irreducible* (in the sense of [CPGSV21]) if the transfer map $\mathcal{E}_A$ has no nontrivial invariant projection, i.e. $\mathcal{E}_A$ is irreducible (Definition 4.23).

Theorem 8.15 (Irreducible block decomposition)**.** *Every MPS tensor A can be decomposed (up to unitary equivalence) into a block-diagonal tensor $\bigoplus_{k=1}^{r} A_k$ where each A_k is irreducible. The block-diagonal tensor generates the same MPV as A.*

Proof. By strong induction on the bond dimension D. If A is already irreducible, take $r = 1$. Otherwise, the transfer map has a PSD fixed point that is not PD (by the Cesàro argument), and Theorem 8.13 splits A into two blocks of strictly smaller bond dimension. Apply the induction hypothesis to each block and concatenate.

Both [PGVWC07] (Theorem 4, "TI canonical form") and [CPGSV21] (§IV.A.1) obtain the block decomposition by the same invariant-projection splitting used here. An alternative route, developed in [Wol12, Thm. 6.14], characterises the fixed-point set of a quantum channel as a $*$-algebra and then applies the Wedderburn–Artin decomposition ([Wol12, Eq. (1.39)]) to read off the block structure directly. The block decomposition is obtained by iterating the invariant-projection splitting and using strong induction on D to ensure termination. □

Theorem 8.16 (Unitary diagonal positive fixed point in TP gauge)**.** *Let A be an irreducible MPS tensor satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and assume $D > 0$. Then there exist a unitary U and a diagonal positive definite matrix Λ such that, for $B^i := U^\dagger A^i U$,*

$$\sum_i (B^i)^\dagger B^i = \mathbb{1} \qquad \textit{and} \qquad \mathcal{E}_B(\Lambda) = \Lambda.$$

Proof. The TP hypothesis makes $\mathcal{E}_A$ a channel, so it has a nonzero PSD fixed point. The irreducible tensor condition implies that $\mathcal{E}_A$ is irreducible as a CP map, and Theorem 5.3 upgrades the fixed point to a positive definite one. The spectral theorem diagonalizes this matrix by a unitary conjugation, and unitary conjugation preserves the TP normalization. □

8.4.1 Burnside bridge: irreducibility and spanning

The irreducible-tensor predicate (Definition 8.14) is formulated in terms of invariant *projections*. For Burnside-style arguments it is more natural to work with invariant *subspaces* of $\mathbb{C}^D$ under the Kraus operators.

Definition 8.17 (Algebra span)**.** For an MPS tensor A, let $\mathrm{algSpan}(A)$ be the unital $\mathbb{C}$-subalgebra of $M_D(\mathbb{C})$ generated by the matrices $\{A^i\}$:

$$\mathrm{algSpan}(A) = \mathbb{C}\langle A^i : i = 0, \dots, d{-}1\rangle \subseteq M_D(\mathbb{C}).$$

Definition 8.18 (Invariant submodule)**.** A subspace $W \leq \mathbb{C}^D$ is *A-invariant* if $A^i W \subseteq W$ for every i.

Definition 8.19 (Irreducible action)**.** The Kraus operators of A act *irreducibly* on $\mathbb{C}^D$ if every A-invariant subspace is trivial, i.e. equal to 0 or to $\mathbb{C}^D$.

Theorem 8.20 (Irreducible tensor implies irreducible action)**.** *If A is irreducible as a tensor (Definition 8.14), then the Kraus operators act irreducibly on $\mathbb{C}^D$ (Definition 8.19).*

Proof. Let $W \leq \mathbb{C}^D$ be an A-invariant subspace. In finite dimension the orthogonal projection P onto W exists. Invariance of W implies $(\mathbb{1} - P)A^i P = 0$ for all i. Thus P is a nontrivial invariant projection unless $W = 0$ or $W = \mathbb{C}^D$, contradicting irreducibility of the tensor. $\square$

Theorem 8.21 (Burnside's theorem for Kraus algebras)**.** *Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then the generated unital subalgebra is the full matrix algebra:*

$$\mathrm{algSpan}(A) = M_D(\mathbb{C}).$$

This is the complex finite-dimensional case of Burnside's theorem (equivalently, Jacobson's density theorem); see [Jac09].

Proof. Transport $\mathrm{algSpan}(A)$ along the algebra equivalence $M_D(\mathbb{C}) \simeq_{\mathbb{C}} \mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$. The irreducible-action hypothesis makes $\mathbb{C}^D$ a simple module for this algebra. Over the algebraically closed field $\mathbb{C}$, Schur's lemma identifies the endomorphisms commuting with the action with scalars, and Jacobson density then forces the action map to be surjective onto all of $\mathrm{End}_{\mathbb{C}}(\mathbb{C}^D)$. $\square$

Theorem 8.22 (Irreducible action implies eventual cumulative spanning)**.** *Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then there exists N such that $T_N(A) = M_D(\mathbb{C})$.*

Proof. By Theorem 8.21, $\mathrm{algSpan}(A) = M_D(\mathbb{C})$. Every element of the algebra span is a linear combination of word products, hence belongs to $T_n(A)$ for some n. Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \leq T_1(A) \leq \cdots$ stabilizes, so one can choose a uniform N with $T_N(A) = M_D(\mathbb{C})$. $\square$

8.5 Proportional single-block theorem

Theorem 8.23 (Proportional MPV implies gauge-phase equivalence)**.** *Let A and B be injective MPS tensors satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If A and B generate proportional MPV families ($V^{(N)}(A)_\sigma = c_N V^{(N)}(B)_\sigma$ for some $c_N \in \mathbb{C}$) and $O_{AA}(N) \to 1$, $O_{BB}(N) \to 1$, then A and B are gauge-phase equivalent.*

Proof. By contradiction. If A and B are not gauge-phase equivalent, then $O_{AB}(N) \to 0$ (Theorem 6.17). From the proportionality hypothesis, $O_{AB}(N) = c_N O_{BB}(N)$. Since $O_{BB}(N) \to 1$, it follows that $c_N \to 0$. But $O_{AA}(N) = |c_N|^2 O_{BB}(N) \to 0$, contradicting $O_{AA}(N) \to 1$.

In [PGVWC07], the proportional case is handled as part of the canonical form argument. Thus the proportional case is obtained directly from the spectral gap in Chapter 6. $\square$

8.6 Canonical form from peripheral primitivity

The canonical form predicate (Definition 9.8) includes an explicit hypothesis that the self-overlap $O_{A_k A_k}(N) \to 1$ for each block. The following theorem derives this from per-block peripheral-spectrum primitivity, removing it as an independent assumption.

Theorem 8.24 (Canonical form from peripheral primitivity)**.** *Let $(\mu_k, A_k)_{k=1}^r$ be a collection of block tensors and scaling factors satisfying:*

1. each A_k is injective,

2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,

3. $|\mu_1| > \cdots > |\mu_r|$ (strictly decreasing),

4. all $\mu_k \neq 0$,

5. each transfer map $\mathcal{E}_{A_k}$ is primitive, i.e. has peripheral spectrum $\{1\}$.

Then (μ_k, A_k) satisfies the canonical form predicate (Definition 9.8).

Proof. Each block is injective and TP-normalized. Peripheral-spectrum primitivity therefore implies the spectral-gap primitive condition used in Chapter 6, and hence the required self-overlap convergence. This supplies the final field in Definition 9.8. □

Remark 8.25. Theorem 8.24 shows that the overlap convergence hypothesis in the canonical form predicate is redundant once peripheral-spectrum primitivity is known. The argument factors through the spectral-gap formulation of Chapter 6: peripheral-spectrum primitivity is first converted to that spectral-gap condition, and the overlap limit is then deduced from it. In the paper [CPGSV21], primitivity is obtained from irreducibility plus periodicity removal by blocking to period one. Theorem 8.26 is precisely this periodicity-removal step under irreducibility and TP normalization.

Theorem 8.26 (Blocking yields a peripheral-spectrum primitive transfer map)**.** *Let A be an irreducible MPS tensor with $D > 0$. Assume that A satisfies the TP normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then there exists a blocking length $p > 0$ such that the transfer map $\mathcal{E}_{A^{[p]}}$ of the blocked tensor $A^{[p]}$ is primitive, i.e. its peripheral spectrum is $\{1\}$.*

Proof. Pass to the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$. The TP normalization of A makes K unital, and irreducibility of A gives irreducibility of the associated Kraus map. The TP fixed-point theorem for $\mathcal{E}_A$ provides a positive definite fixed point for the adjoint map of K, so Theorem 4.45 shows that every peripheral eigenvalue is a root of unity. Since peripheral$(\mathcal{E}_A)$ is finite (Theorem 4.37), Lemma 4.41 gives a common exponent p such that $\mu^p = 1$ for all $\mu \in$ peripheral$(\mathcal{E}_A)$. Lemma 4.42 then implies peripheral$(\mathcal{E}_A^p) = \{1\}$. Finally, blocking corresponds to taking a power of the transfer map, so $\mathcal{E}_{A^{[p]}}$ is primitive. □

Remark 8.27. Theorem 8.26 is the “periodicity removal by blocking” step in [CPGSV17, Appendix A] for an irreducible block already placed in TP gauge. The later existence theorems simply reuse this step.

8.7 Normal canonical form

Definition 8.28 (Normal canonical form predicate)**.** A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *normal canonical form predicate* if:

1. each A_k is irreducible,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. each transfer map $\mathcal{E}_{A_k}$ is primitive (equivalently, has peripheral spectrum $\{1\}$),
4. the moduli are strictly decreasing: $|\mu_1| > |\mu_2| > \cdots > |\mu_r|$,

5. all $\mu_k \neq 0$,

6. all bond dimensions are positive.

Here "normal" is used in the sense of [CPGSV17]: irreducible blocks with trivial peripheral spectrum after TP normalization. It is not the same as Definition 2.19, which means eventual block injectivity.

Theorem 8.29 (Self-overlap is derived in normal canonical form)**.** *If (μ_k, A_k) satisfies the normal canonical form predicate, then $O_{A_k A_k}(N) \to 1$ for every block k.*

Proof. Peripheral-spectrum primitivity is part of the data. Together with irreducibility and TP normalization, it yields the spectral-gap primitive theorem needed for overlap convergence, so the self-overlap condition is derived rather than stored as an extra field. □

Theorem 8.30 (Modulus-one eigenvalue rigidity for irreducible TP blocks)**.** *Let A and B be irreducible MPS tensors satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. If the mixed transfer map has spectral radius at least 1 (equivalently, it has a peripheral eigenvalue of modulus 1), then A and B are gauge-phase equivalent.*

Proof. A modulus-one eigenvector for the mixed transfer map saturates the channel Schwarz inequality. Irreducibility then upgrades this eigenvector to an intertwiner between the two Kraus families, which is exactly gauge-phase equivalence. □

8.8 Steps toward canonical form existence

This section assembles the components of the canonical-form construction following [CPGSV17, §2.3 and Appendix A].

Theorem 8.31 (Irreducible block decomposition)**.** *Every MPS tensor A generates the same MPV family as a block-diagonal tensor $\bigoplus_{k=1}^{r} A_k$ with each block irreducible.*

Proof. Apply Theorem 8.15. □

Theorem 8.32 (CFII data for irreducible TP blocks)**.** *Let A be an irreducible MPS tensor already in TP gauge ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) with $D > 0$. Then there exist a unitary U and a diagonal positive definite matrix Λ such that:*

1. *$B^i := U^\dagger A^i U$ is TP,*

2. *$\mathcal{E}_B(\Lambda) = \Lambda$,*

3. *Λ is diagonal and positive definite,*

4. *A and B generate the same MPV family.*

This is the "Canonical Form II" (CFII) step of [CPGSV17, Appendix A]. The unitary conjugation occurs only after TP normalization has already been achieved; it is a second step inside TP gauge, not a unitary normalization of an arbitrary irreducible block.

Proof. Apply Theorem 8.16 to obtain the unitary and the diagonal PD fixed point. Unitary conjugation preserves trace-preservation and the MPV family (Theorem 8.9). □

Theorem 8.33 (CFII continuation after irreducible block decomposition)**.** *For any MPS tensor A, there exist irreducible blocks $(A_k)_{k=1}^r$ that generate the same MPV family as A. Moreover, for each block k, if that block has already been placed in TP gauge and $D_k > 0$, one obtains CFII data (unitary, diagonal PD fixed point, MPV preservation). The conclusion is CFII data; the normal-canonical-form packaging requires additional downstream steps.*

Proof. Combine Theorems 8.31 and 8.32. □

Theorem 8.34 (Normal canonical form from a primitive block decomposition)**.** *Suppose A generates the same MPV family as a weighted block-diagonal tensor $\bigoplus_k \mu_k A_k$, and assume that:*

1. *each A_k is irreducible,*
2. *each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,*
3. *each transfer map $\mathcal{E}_{A_k}$ is primitive (equivalently, has peripheral spectrum $\{1\}$),*
4. *the moduli $|\mu_k|$ are pairwise distinct,*
5. *all $\mu_k \neq 0$,*
6. *all bond dimensions are positive.*

Then there exist a blocking length $p > 0$, weights (ν_k), and blocked tensors (B_k) such that $A^{[p]}$ generates the same MPV family as $\bigoplus_k \nu_k B_k$, and (ν_k, B_k) satisfies the normal canonical form predicate.

Proof. The theorem first performs a common blocking of the already primitive blocks; in the present statement this blocking is trivial, so one may take $p = 1$. It then reorders the blocks by decreasing weight norm. The resulting irreducibility, TP, primitivity, nonzero-weight, and positive-dimension data are exactly those required by Definition 8.28. □

8.9 Open directions

Remark 8.35 (Canonical form reduction: remaining steps). The Perron–Frobenius (Chapter 5), TP-gauge, CFII, periodicity-removal, and normal-canonical-form packaging steps are in place. The upstream passage from an arbitrary tensor to the primitive weighted block data required by Theorem 8.34 remains open.

On the block-injective side, turning blocked primitive TP data into injective blocks and establishing eventual block injectivity are the open steps toward Definition 9.8.

On the normal-canonical side, Theorem 8.34 provides the downstream packaging once primitive weighted blocks with positive dimensions and pairwise distinct nonzero weight moduli are given. Theorem 8.33 gives the CFII continuation after irreducible block decomposition; the end-to-end construction from arbitrary tensors requires additional work.

Chapter 9

Block Permutation and Separation

This chapter establishes the algebraic structure of automorphisms of products of matrix algebras and uses it to prove that the same-MPV condition on block-diagonal tensors descends to per-block gauge equivalence. The approach follows [PGVWC07] and [CPGSV21].

9.1 Block permutation algebra

Theorem 9.1 (Ring isomorphisms permute block ideals)**.** *Let $\{R_k\}_{k=1}^r$ be simple rings. Every ring isomorphism $T : \prod_k R_k \xrightarrow{\sim} \prod_k R_k$ permutes the block ideals: there exists a permutation σ such that T maps the k-th block ideal to the $\sigma(k)$-th block ideal.*

Proof. Each block ideal $I_k = \{x : x_j = 0 \text{ for } j \neq k\}$ is an atom in the lattice of two-sided ideals (since R_k is simple). A ring isomorphism preserves the ideal lattice and hence permutes the atoms. Injectivity of the induced map on atoms gives the permutation σ. □

Theorem 9.2 (Dimension preservation)**.** *If T is an algebra automorphism of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$, and σ is the induced permutation, then $D_{\sigma(k)} = D_k$ for all k.*

Proof. The restriction of T to the k-th block ideal induces a $\mathbb{C}$-algebra isomorphism $M_{D_k}(\mathbb{C}) \cong M_{D_{\sigma(k)}}(\mathbb{C})$, which forces $D_k = D_{\sigma(k)}$ (by comparing dimensions as $\mathbb{C}$-vector spaces). □

Theorem 9.3 (Decomposition of automorphisms of $\prod M_{D_k}(\mathbb{C})$)**.** *Every $\mathbb{C}$-algebra automorphism of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$ decomposes as a block permutation σ followed by per-block inner automorphisms: there exist $\sigma \in S_r$ with $D_{\sigma(k)} = D_k$ and invertible matrices $X_k \in \mathrm{GL}_{D_k}(\mathbb{C})$ such that $T(M)_k = X_k M_{\sigma(k)} X_k^{-1}$.*

Proof. Theorem 9.1 gives the permutation σ. Theorem 9.2 gives $D_{\sigma(k)} = D_k$. The restriction of T to each block is a $\mathbb{C}$-algebra automorphism of $M_{D_k}(\mathbb{C})$, which is inner by Skolem–Noether (Theorem 3.9).

This is a standard result in the theory of semisimple algebras; see [CPGSV21, §IV.A] for the MPS context. □

9.2 Pi-algebra linear extension

Definition 9.4 (Per-block linear extension)**.** Given injective tensors A_k, B_k of bond dimension D_k with the same MPV for each block k, the *per-block linear extension* T_k is the unique linear map $M_{D_k}(\mathbb{C}) \to M_{D_k}(\mathbb{C})$ satisfying $T_k(A_k^i) = B_k^i$ for all i. It exists (and is unique) because $\{A_k^i\}$ spans $M_{D_k}(\mathbb{C})$ by injectivity.

Definition 9.5 (Product algebra automorphism)**.** The per-block linear extensions T_k are $\mathbb{C}$-algebra homomorphisms, and assembling them blockwise gives a single $\mathbb{C}$-algebra automorphism $T : \prod_k M_{D_k}(\mathbb{C}) \xrightarrow{\sim} \prod_k M_{D_k}(\mathbb{C})$.

Theorem 9.6 (Per-block linear extensions assemble)**.** *Let A_k and B_k be injective tensors of bond dimension D_k for $k = 1, \dots, r$, and suppose the per-block same-MPV condition holds. Then the per-block linear extensions $T_k : A_k^i \mapsto B_k^i$ assemble into an algebra automorphism of $\prod_k M_{D_k}(\mathbb{C})$, which decomposes as a permutation σ with $D_{\sigma(k)} = D_k$ and per-block conjugation matrices X_k.*

Proof. Each T_k exists uniquely and is multiplicative by Theorems 3.6 and 3.7. Promotion to algebra homomorphisms (Lemma 3.10) and assembly into the product gives an algebra automorphism. Theorem 9.3 decomposes it into permutation + per-block inner automorphisms. □

Theorem 9.7 (Per-block same MPV gives global gauge equivalence)**.** *If all block tensors A_k are injective and per-block same-MPV holds, then each pair (A_k, B_k) is gauge equivalent, and the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are globally gauge equivalent.*

Proof. Per-block gauge equivalence follows from the single-block FT (Theorem 3.11) applied to each block. Global gauge equivalence follows by assembling the per-block gauge matrices into a block-diagonal matrix (Theorem 8.4). □

9.3 Block separation

Definition 9.8 (Canonical form predicate)**.** A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *canonical form predicate* if:

1. each A_k is injective,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. the moduli are strictly decreasing: $|\mu_1| > |\mu_2| > \cdots > |\mu_r|$,
4. all $\mu_k \neq 0$,
5. the self-overlap converges: $O_{A_k A_k}(N) \to 1$ for each k.

Only this one-sided normalization is part of the predicate; no unital condition is assumed here. Condition (5) is a primitivity hypothesis: for a primitive channel ([Wol12, Thm. 6.7, item 3]), $T^n(\rho) \to \rho_\infty$ for every initial state, which forces the self-overlap to converge to 1. See also [Wol12, Thm. 6.8] for the completely-positive characterisation and Chapter 6 for the spectral-gap proof.

Remark 9.9. In the literature [CPGSV21], condition (5) is derived from primitivity of the transfer map (peripheral spectrum = $\{1\}$). Theorem 8.24 gives this implication: per-block primitivity implies overlap convergence, so condition (5) is redundant when primitivity holds. The canonical form predicate retains condition (5) explicitly because the overlap limit may also be assumed directly.

Remark 9.10 (Normal-canonical companion). Definition 9.8 is the block-injective predicate. The normal-canonical route instead uses Definition 8.28: injectivity and the explicit self-overlap hypothesis are replaced by irreducibility, TP normalization, and primitive transfer maps. Theorem 8.29 then supplies the self-overlap limit automatically.

Theorem 9.11 (Vandermonde separation)**.** *If $\mu_1, \dots, \mu_r$ are pairwise distinct complex numbers and $\sum_{k=1}^{r} c_k \mu_k^N = 0$ for $N = 0, 1, \dots, r-1$, then $c_k = 0$ for all k.*

Proof. The coefficient vector c satisfies $Vc = 0$ where V is the Vandermonde matrix of the μ_k. Injectivity of μ makes V invertible. $\square$

Lemma 9.12 (Block separation core)**.** *Let $((\mu_k), (A_k))$ satisfy the canonical form predicate, and let (B_k) be injective tensors satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. If $\sum_k \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0$ for all N, σ, then each pair (A_k, B_k) generates the same MPV family.*

Proof. By induction on the number of blocks r. In the inductive step, take overlaps with the leading block A_0; cross-block contributions from $k \geq 1$ decay exponentially (by the spectral gap), leaving $O_{A_0 A_0}(N) - O_{A_0 B_0}(N) \to 0$. The proportional FT (Theorem 8.23) gives $A_0 \simeq B_0$. Peeling off the leading block and dividing by μ_0^N, the induction hypothesis applies. $\square$

Theorem 9.13 (Block separation from canonical form)**.** *Let $((\mu_k), (A_k))$ satisfy the canonical form predicate, and let (B_k) be injective tensors satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) generates the same MPV family.*

Proof. Apply Lemma 9.12 to the identity obtained from the equality of the two block-diagonal MPV families. The canonical form predicate supplies injectivity, TP normalization, strict nonzero weights, and self-overlap convergence for (A_k), while the hypotheses on (B_k) supply the comparison injectivity and TP normalization. $\square$

Theorem 9.14 (Block separation from normal canonical form)**.** *Let $((\mu_k), (A_k))$ satisfy the normal canonical form predicate, and let (B_k) be irreducible tensors of the same bond dimensions, satisfying the same TP normalization $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) generates the same MPV family.*

Proof. The proof follows the same peeling argument as Theorem 9.13. Theorem 8.29 replaces the explicit self-overlap hypothesis on the leading block, and Theorem 8.30 identifies the leading comparison block from the resulting nondecaying mixed overlap. Since both block families already use the same dimension function, the conclusion is direct per-block same MPV. $\square$

Theorem 9.15 (Canonical form fundamental theorem, same-structure version)**.** *Under the hypotheses of Theorem 9.13, each pair (A_k, B_k) is gauge equivalent. Hence the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

Proof. Theorem 9.13 gives per-block same MPV. Theorem 9.7 converts this to per-block gauge equivalence and then to global gauge equivalence of the block-diagonal tensors. $\square$

Chapter 10

Basis Normal Tensors

This chapter introduces *Basis Normal Tensors* (BNT), the canonical-form variants that separate distinct blocks, the cross-overlap decay theorem, and the overlap-based permutation rigidity arguments used in the final assembly. The coefficient-convergence and Newton–Girard auxiliary results are also collected here. The main references are [PGVWC07] and [CPGSV21].

10.1 Basis normal tensors

Definition 10.1 (Basis Normal Tensor (BNT))**.** A *basis normal tensor* (BNT) decomposition for a total tensor A_{tot} consists of g block tensors $(A_1, \dots, A_g)$ with bond dimensions $(D_1, \dots, D_g)$ such that:

1. each A_j is normal (Definition 2.19),
2. at each system size N, the MPV of A_{tot} is a linear combination of the block MPVs: $|V^{(N)}(A_{\mathrm{tot}})\rangle = \sum_j c_j |V^{(N)}(A_j)\rangle$,
3. for sufficiently large N, the block MPV vectors $\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g}$ are linearly independent.

This is [CPGSV21, Definition 4.2].

Definition 10.2 (Canonical form with BNT separation)**.** A *canonical form with BNT separation* extends the canonical form (Definition 9.8) with the additional requirement that distinct blocks are *not* gauge-phase equivalent: for $j \neq k$ with $D_j = D_k$, A_j and A_k are not gauge-phase equivalent.

Definition 10.3 (Normal canonical form with BNT separation)**.** A *normal canonical form with BNT separation* extends the normal canonical form predicate (Definition 8.28) with the same non-equivalence condition as Definition 10.2: for $j \neq k$ with $D_j = D_k$, the blocks A_j and A_k are not gauge-phase equivalent.

Theorem 10.4 (Cross-overlap decay for distinct CF-BNT blocks)**.** *If $((\mu_k), (A_k))$ satisfies the CF-BNT predicate and $j \neq k$, then $O_{A_j A_k}(N) \to 0$.*

Proof. If $D_j \neq D_k$, the rectangular spectral gap theorem (Theorem 6.15) gives $O_{A_j A_k}(N) \to 0$. If $D_j = D_k$, the blocks are not gauge-phase equivalent (by the CF-BNT separation condition), so the spectral-gap theorem (Theorem 6.17) gives decay to 0. □

Theorem 10.5 (CF-BNT yields BNT). *If $((\mu_k), (A_k))$ satisfies the CF-BNT predicate, then the block tensors form a BNT decomposition for the block-diagonal tensor $\bigoplus_k \mu_k A_k$.*

Proof. Each block is injective, hence 1-block injective and therefore normal in the sense of Definition 2.19. The MPV decomposition (Theorem 2.24) gives $V^{(N)}(A_{\text{tot}})_\sigma = \sum_k \mu_k^N V^{(N)}(A_k)_\sigma$. The overlap matrix $G_{jk}(N) = O_{A_j A_k}(N)$ converges to the identity: diagonal entries tend to 1 by the self-overlap hypothesis, and off-diagonal entries tend to 0 by Theorem 10.4. Hence $G(N)$ is eventually invertible, so the block MPV vectors are eventually linearly independent. $\square$

Remark 10.6 (The normality gap for normal-CF-BNT). Definition 10.3 uses "normal" in the normal-canonical sense, not the eventual block-injective predicate of Definition 2.19. Therefore normal-CF-BNT data do not by themselves yield the BNT predicate of Definition 10.1; one still needs eventual block injectivity for each block.

10.2 BNT permutation rigidity

Theorem 10.7 (BNT permutation — proportional MPV case). *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be injective families satisfying the TP normalization $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ and $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$, whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. Assume moreover that there exist MPS tensors A_{tot} and B_{tot}, coefficient arrays $a_{N,j}$ and $b_{N,k}$, and a sequence c_N with nonzero limits a_j^∞, b_k^∞, c^∞ such that for every N and σ,*

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^{g_A} a_{N,j} V^{(N)}(A_j)_\sigma, \qquad V^{(N)}(B_{\text{tot}})_\sigma = \sum_{k=1}^{g_B} b_{N,k} V^{(N)}(B_k)_\sigma,$$

with $a_{N,j} \to a_j^\infty \neq 0$ and $b_{N,k} \to b_k^\infty \neq 0$, and also

$$V^{(N)}(A_{\text{tot}})_\sigma = c_N V^{(N)}(B_{\text{tot}})_\sigma$$

with $c_N \to c^\infty \neq 0$. Then $g_A = g_B$, and there is a permutation $\sigma \in S_g$ with $D_{\sigma(k)} = D_k$ and $A_{\sigma(k)}$ gauge-phase equivalent to B_k for each k.

Proof. From the proportional MPV hypothesis and convergent coefficients, one extracts the limit of each mixed overlap $O_{A_j B_k}(N)$. The overlap-orthonormal hypotheses force every row and column of the limiting mixed-overlap matrix to contain at most one nonzero entry, hence define a permutation. A nonzero mixed overlap cannot decay, so the spectral gap upgrades it to gauge-phase equivalence of the matched blocks.

This is the overlap-and-decomposition form of the permutation step corresponding to [CPGSV21, Theorem 4.1]. $\square$

Theorem 10.8 (BNT permutation — overlap-orthonormal case). *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be two families of injective tensors satisfying the TP normalization $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ and $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$, whose self-overlaps converge to 1 and whose cross-overlaps (within each family) converge to 0. If they span the same MPV subspace at every system size, then $g_A = g_B$ and the blocks are related by a permutation and gauge-phase equivalence.*

Proof. The overlap matrix $O_{A_j B_k}(N)$ converges to a matrix with at most one nonzero entry per row/column (by the spectral gap). The overlap orthonormality hypotheses are assumed directly, rather than derived from proportional MPVs together with coefficient convergence. The irreducible-TP variant uses the same matching argument with the irreducible-TP overlap decay theorems in place of the injective ones. $\square$

10.3 Coefficient convergence

Theorem 10.9 (Coefficient ratio decay)**.** *Under the canonical form predicate, for* $k \geq 1$, $(\mu_k/\mu_0)^N \to 0$ *as* $N \to \infty$.

Proof. Since $|\mu_k| < |\mu_0|$ (strict ordering of moduli), $|\mu_k/\mu_0| < 1$, so the geometric sequence converges to 0. □

Remark 10.10 (Strict-dominance regime). Theorem 10.9 applies when one modulus is strictly dominant: $|\mu_0| > |\mu_k|$ for $k \geq 1$. The oscillatory case $\sum_q \mu_{j,q}^N$ with $|\mu_{j,q}| = 1$ is treated separately in the full paper.

10.4 Newton–Girard identities

Theorem 10.11 (Newton–Girard trace recursion)**.** *If* $A, B \in M_n(\mathbb{C})$ *satisfy* $\mathrm{tr}(A^k) = \mathrm{tr}(B^k)$ *for all* $k \geq 1$*, then* A *and* B *have the same characteristic polynomial.*

Proof. The Newton–Girard identities express the coefficients of the characteristic polynomial recursively in terms of the power-sum traces $\mathrm{tr}(A^k)$. Equal power sums give equal coefficients, hence equal characteristic polynomials. □

Theorem 10.12 (Equal power sums imply equal multisets)**.** *Let* $\alpha, \beta : \{1, \dots, n\} \to \mathbb{C}$*. If* $\sum_i \alpha_i^k = \sum_i \beta_i^k$ *for all* $k \geq 1$*, then the multisets* $\{\alpha_i\}$ *and* $\{\beta_i\}$ *are equal.*

Proof. Construct diagonal matrices $\mathrm{diag}(\alpha)$ and $\mathrm{diag}(\beta)$. The power-sum hypothesis gives $\mathrm{tr}(\mathrm{diag}(\alpha)^k) = \mathrm{tr}(\mathrm{diag}(\beta)^k)$. By Theorem 10.11, the characteristic polynomials agree: $\prod_i (X - \alpha_i) = \prod_i (X - \beta_i)$. Extracting roots gives multiset equality. □

Chapter 11

Full Assembly

This chapter states the assembled Fundamental Theorem results. The proportional-MPV theorems take explicit coefficient-convergence input in both the block-injective CF-BNT setting and the normal-canonical CF-BNT setting. The equal-MPV theorem is the common-block-structure specialization. The results combine the canonical form reduction (Chapter 8), block separation (Chapter 9), spectral gap (Chapter 6), and the BNT permutation arguments of Chapter 10.

11.1 Proportional Fundamental Theorem

Theorem 11.1 (Fundamental Theorem — proportional MPV case)**.** *Let A_{tot} and B_{tot} be MPS tensors equipped with CF-BNT families $(\mu_j^A, A_j)_{j=1}^{r_A}$ and $(\mu_k^B, B_k)_{k=1}^{r_B}$. Assume one is given coefficient arrays $a_{N,j}$, $b_{N,k}$ and nonzero limits a_j^∞, b_k^∞, c^∞ such that*

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = \sum_{j=1}^{r_A} a_{N,j} V^{(N)}(A_j)_\sigma, \qquad V^{(N)}(B_{\mathrm{tot}})_\sigma = \sum_{k=1}^{r_B} b_{N,k} V^{(N)}(B_k)_\sigma,$$

with $a_{N,j} \to a_j^\infty \neq 0$ and $b_{N,k} \to b_k^\infty \neq 0$, and suppose also that

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = c_N V^{(N)}(B_{\mathrm{tot}})_\sigma$$

for a sequence $c_N \to c^\infty \neq 0$. Then $r_A = r_B$, and there is a permutation π such that $D_j^A = D_{\pi(j)}^B$ and A_j is gauge-phase equivalent to $B_{\pi(j)}$ for each j.

Proof. Apply Theorem 10.7 to the two CF-BNT decompositions. The CF-BNT hypotheses provide injectivity, TP normalization, diagonal overlap limits, and off-diagonal decay via Theorem 10.4. With the supplied decomposition identities and convergence hypotheses, Theorem 10.7 yields the equality of block counts, the permutation, and the per-block gauge-phase equivalence. □

Theorem 11.2 (Fundamental Theorem — proportional MPV case, NT route)**.** *Let A_{tot} and B_{tot} be MPS tensors equipped with normal-CF-BNT families $(\mu_j^A, A_j)_{j=1}^{r_A}$ and $(\mu_k^B, B_k)_{k=1}^{r_B}$. Assume one is given coefficient arrays $a_{N,j}$, $b_{N,k}$ and nonzero limits a_j^∞, b_k^∞, c^∞ such that*

$$V^{(N)}(A_{\mathrm{tot}})_\sigma = \sum_{j=1}^{r_A} a_{N,j} V^{(N)}(A_j)_\sigma, \qquad V^{(N)}(B_{\mathrm{tot}})_\sigma = \sum_{k=1}^{r_B} b_{N,k} V^{(N)}(B_k)_\sigma,$$

with $a_{N,j} \to a_j^\infty \neq 0$ *and* $b_{N,k} \to b_k^\infty \neq 0$*, and suppose also that*

$$V^{(N)}(A_{\text{tot}})_\sigma = c_N V^{(N)}(B_{\text{tot}})_\sigma$$

for a sequence $c_N \to c^\infty \neq 0$*. Then* $r_A = r_B$*, and there is a permutation* π *such that* $D_j^A = D_{\pi(j)}^B$ *and* A_j *is gauge-phase equivalent to* $B_{\pi(j)}$ *for each* j*.*

Proof. Normal-CF-BNT data provide irreducibility, TP normalization, primitive self-overlap convergence via Theorem 8.29, and the same off-diagonal decay mechanism as in the injective CF-BNT case. The same matching argument yields the block permutation and gauge-phase equivalence. Remark 10.6 explains why the argument stops at this separated form. □

11.2 Equal-MPV Fundamental Theorem

Theorem 11.3 (Fundamental Theorem — equal MPV case)**.** *Fix a common block structure: the same number of blocks* r*, the same bond dimensions* $(D_k)_{k=1}^r$*, and the same weights* $(\mu_k)_{k=1}^r$*. Let* (μ_k, A_k) *and* (μ_k, B_k) *be two CF-BNT families on this common data. If the block-diagonal tensors* $\bigoplus_k \mu_k A_k$ *and* $\bigoplus_k \mu_k B_k$ *generate the same MPV family, then each pair* (A_k, B_k) *is gauge equivalent, and hence the assembled block-diagonal tensors are gauge equivalent.*

Proof. Forget the extra BNT-separation field and retain only the underlying canonical-form data. The canonical-form fundamental theorem (Theorem 9.15) then applies directly to this common (r, D_k, μ_k) structure, yielding per-block gauge equivalence and hence global gauge equivalence. □

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